WSQ · SOFTWARE DESIGN

# Build ASP.NET Web Apps with Vibe Coding



**Course Code: TGS-2023040477**

Software Design · ICT-DES-3005-1.1

Trainer: Dr. Alfred Ang

Version 6.1 · 12 September 2026

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

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# Digital Attendance (Mandatory)

1

## AM attendance

Scan the SSG QR code shown by the trainer.

2

## PM attendance

Complete the second attendance event after lunch.

3

## Assessment attendance

Complete the dedicated assessment attendance.

4

## Evidence

Minimum 75% attendance and competent assessment are funding conditions.


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# About the Trainer




## General Trainer template

Complete before delivery

**NAME** _____

**TITLE / DESIGNATION** _____

**QUALIFICATIONS** _____

**AREAS OF EXPERTISE** _____

**TRAINING & INDUSTRY EXPERIENCE** _____

**CONTACT** _____


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# About the Trainer




## Dr. Alfred Ang

Principal Trainer · software and AI-enabled development

| | |
|---|---|
| NAME | Dr. Alfred Ang |
| TITLE / DESIGNATION | Principal Trainer |
| QUALIFICATIONS | Doctorate · industry certifications |
| AREAS OF EXPERTISE | ASP.NET · software design · AI coding |
| TRAINING & INDUSTRY EXPERIENCE | Adult education and industry practice |
| CONTACT | enquiry@tertiaryinfotech.com |


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LEARNER INTRODUCTION

# Let's Know Each Other

1

## Name + role

Share your current work context.

2

## Web experience

Name one app or API you have used.

3

## ASP.NET baseline

Share your .NET/C# exposure.

4

## Learning goal

Name one CRM outcome you want to build.


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LEARNING ENVIRONMENT

# Ground Rules

1 **Participate**

Ask, explain and test actively.

2 **Respect**

One conversation; challenge ideas, not people.

3 **Be punctual**

Return from breaks on time.

4 **Protect data**

Use only provided mock CRM records.

5 **Verify AI**

Inspect every generated diff and dependency.

6 **Attend**

Complete AM, PM and assessment attendance.


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COURSE ADMINISTRATION · 09:00–18:00

# Lesson Plan · Day 1 · Part 1/2

| | | |
|---|---|---|
| **09:00–09:30** | Administration, attendance, introductions and assessment briefing | Facilitation |
| **09:30–10:45** | Topic 1 — ASP.NET Core runtime and component design | Lecture, demonstration, questioning |
| **10:45–10:55** | Morning break | Break |
| **10:55–12:30** | Activities 1–2 — requirements, scaffold and controlled vibe change | Practical |
| **12:30–13:30** | Lunch | Break |
| **13:30–14:50** | Topic 2 — accessible UI and organisational practice | Lecture, demonstration, questioning |
| **14:50–15:40** | Activities 3–4 — accessible shell and audit | Practical |

Continued on the next schedule slide.


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COURSE ADMINISTRATION · 09:00–18:00

# Lesson Plan · Day 1 · Part 2/2

| | | |
|---|---|---|
| **15:40–15:50** | Afternoon break | Break |
| **15:50–17:00** | Topic 3 — Razor Pages controls, binding and persistence | Lecture and demonstration |
| **17:00–17:50** | Activity 5 — create and list with Razor Pages | Practical |
| **17:50–18:00** | Day 1 recap and PM attendance | Reflection |

Tea breaks sit within facilitated time; lunch is excluded. Final assessment runs Day 2, 16:00–18:00.


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COURSE ADMINISTRATION · 09:00–18:00

# Lesson Plan · Day 2 · Part 1/2

| | | |
|---|---|---|
| **09:00–09:15** | Day 1 recap and AM attendance | Review |
| **09:15–10:05** | Topic 3 — validation, Tag Helpers and antiforgery | Lecture and demonstration |
| **10:05–10:55** | Activity 6 — validation and antiforgery | Practical |
| **10:55–11:05** | Morning break | Break |
| **11:05–12:10** | Topic 4 — controller APIs and HTTP contracts | Lecture, demonstration, questioning |
| **12:10–13:00** | Activity 7 — controller API | Practical |
| **13:00–14:00** | Lunch | Break |

Continued on the next schedule slide.


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COURSE ADMINISTRATION · 09:00–18:00

# Lesson Plan · Day 2 · Part 2/2

| | | |
|---|---|---|
| **14:00–14:55** | Activity 8 — SQLite, OpenAPI and integration tests | Practical |
| **14:55–15:35** | Topic 5 — hosting, Azure and design evidence | Lecture and demonstration |
| **15:35–15:45** | Afternoon break | Break |
| **15:45–16:00** | Activities 9–10 briefing and course synthesis | Demonstration and reflection |
| **16:00–17:00** | **Written Assessment — Short-Answer Questions (open book)** | Summative assessment |
| **17:00–18:00** | **Practical Performance Assessment (open book)** | Summative assessment |

Tea breaks sit within facilitated time; lunch is excluded. Final assessment runs Day 2, 16:00–18:00.


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# Learning Outcomes

1

## LO1

Determine basic software components using ASP.NET framework to meet functional specifications.

2

## LO2

Utilize ASP.NET framework to develop web apps in adherence to organizational practices.

3

## LO3

Select essential ASP.NET controls and features to meet software design requirements.

4

## LO4

Examine the interoperability and functionality of ASP.NET software components.

5

## LO5

Generate ASP.NET design documentation aligned with user specifications.


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# Skills Framework Alignment

1

## K1 / A1

Requirements and component boundaries · Activities 1–2

2

## K2 / A2

Design tools, accessibility and controlled practice · Activities 3–4

3

## K3 / A3

Razor controls, binding, validation and antiforgery · Activities 5–6

4

## K4 / A4

Controller API, status/schema interoperability · Activities 7–8

5

## K5 / A5

Deployment and traceable design documentation · Activities 9–10


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BEFORE THE ASSESSMENT

# Briefing for Assessment

1 **Open book**

Slides, Learner Guide and approved Activity material only.

2 **Individual evidence**

No discussion, messaging, screenshots or recording.

3 **WA first**

Five short-answer questions map K1–K5; 60 minutes.

4 **PP second**

Five practical tasks map A1–A5; 60 minutes.

5 **Submission**

Upload the required evidence on the LMS.

6 **Clarification**

Assessor may use oral clarification only to close a minor gap.


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SUMMATIVE · INDIVIDUAL · OPEN BOOK

# Assessment and Funding

### 1 Written Assessment

WA-SAQ · 1 hour · K1–K5

### 2 Practical Performance

PP · 1 hour · A1–A5

### 3 Competency

Complete both modes and be assessed Competent.

### 4 Funding condition

At least 75% verified SSG attendance; eligibility and terms apply.


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MANDATORY SEQUENCE

# Assessment Flow




**Courseware and assessment: https://lms-tms.tertiaryinfotech.com/**


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ONE CUMULATIVE LOCAL CRM · 10 SNAPSHOTS

# Access the Hands-On Activities

1

### Option A · clone

git clone https://github.com/tertiarycourses/TGS-2023040477-Build-ASP.NET-Web-Apps-with-Vibe-Coding.git

2

### Option B · ZIP

GitHub · Code · Download ZIP

3

### Run locally

Each Activity has src/LocalCRM plus its own mock-data folder.

4

### Start point

Open activities/README.md and choose Activity 01.

https://github.com/tertiarycourses/TGS-2023040477-Build-ASP.NET-Web-Apps-with-Vibe-Coding


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# Modern ASP.NET Core · Legacy Boundary

### 1 Current course stack

ASP.NET Core 10: Razor Pages, controllers, OpenAPI, EF Core and Kestrel.

### 2 ASP.NET 4.x

Web Forms/MVC 5 depend on .NET Framework/System.Web; historical comparison only.

### 3 Classic ASP

.asp and VBScript are a separate legacy technology; not taught as current Core.

### 4 Source rule

Microsoft Learn current docs outrank secondary legacy tutorials.


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TOPIC 1 · LO1 · K1 / A1

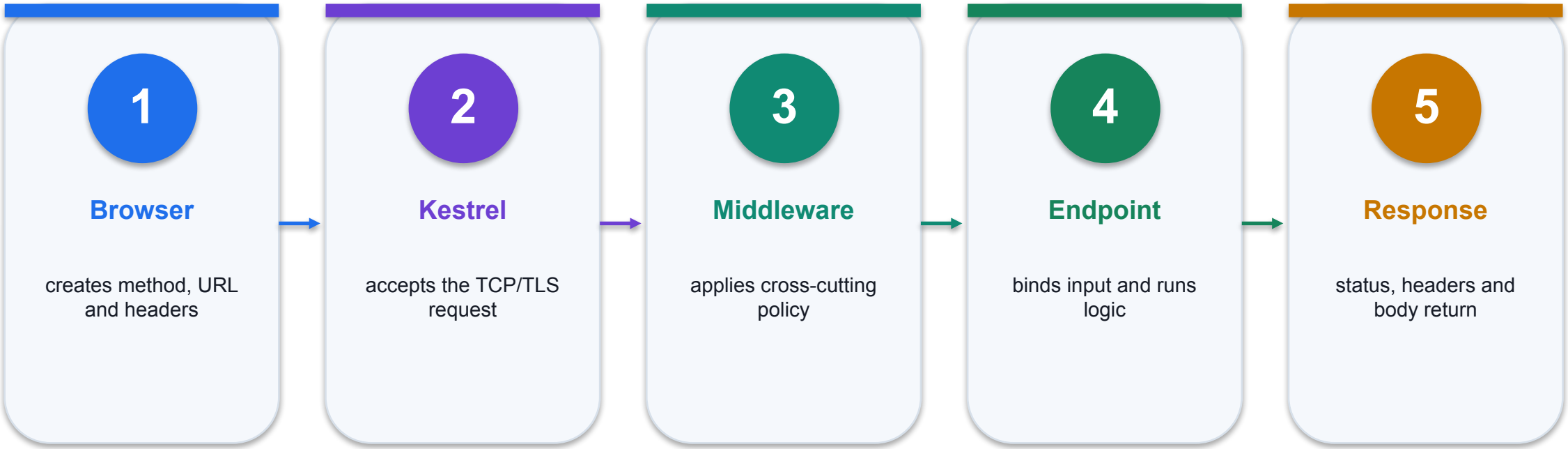
01

# Design Components and the ASP.NET Core Runtime

requirements → component boundaries → request pipeline → verified scaffold


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# HTTP request and response contract




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# HTTP request and response contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  app.MapGet("/health", () => Results.Ok(new { status =
   "ok" }));
```

1 **Method**

GET reads; POST creates; PUT replaces; DELETE removes

2 **Target**

/contacts/42 plus query string

3 **Headers**

Accept, Content-Type, Authorization, trace context

4 **Body**

JSON or form payload; never implied by a GET

K1 / A1

Full procedures: Learner Guide


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# HTTP request and response contract · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Method** | GET reads; POST creates; PUT replaces; DELETE removes | Where is this enforced? |
| **Target** | /contacts/42 plus query string | What is observable? |
| **Headers** | Accept, Content-Type, Authorization, trace context | Who owns the value? |
| **Body** | JSON or form payload; never implied by a GET | What rejects invalid input? |
| **Status** | 2xx success, 4xx caller issue, 5xx server failure | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# HTTP request and response contract · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| 404 | route template did not match |
| 405 | path matched but method did not |
| 415 | Content-Type cannot be read |
| 500 | unhandled endpoint exception |

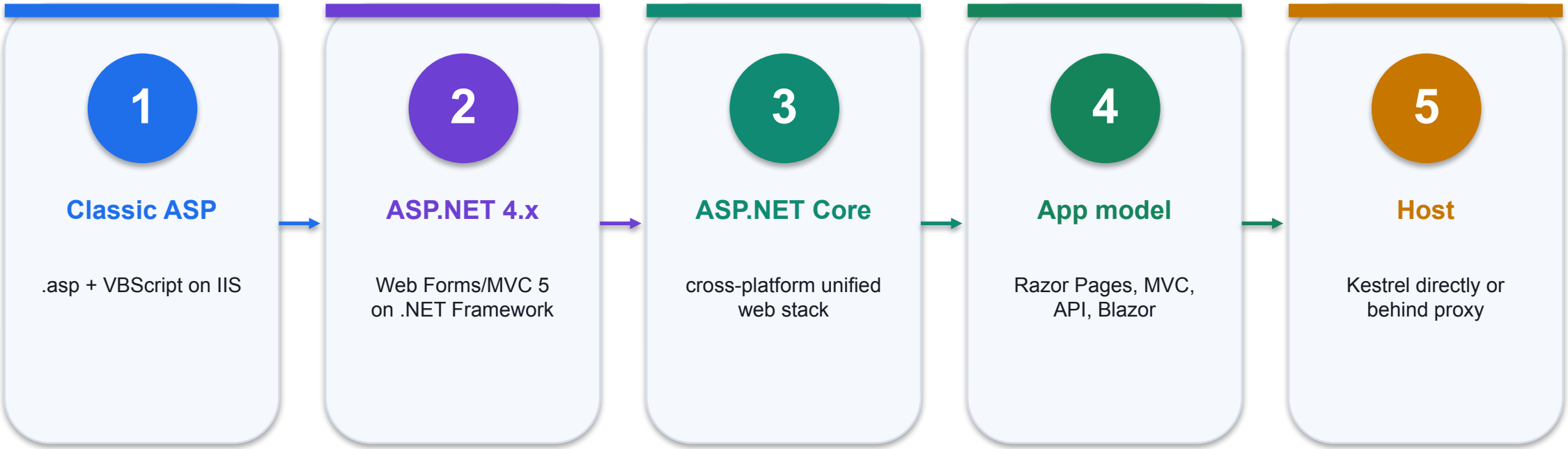
## ACCEPTANCE EVIDENCE

- ✓ curl -i /health returns HTTP 200
- ✓ Content-Type is application/json
- ✓ The response body matches the declared contract

**Record URL · command · status · build SHA**


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# ASP.NET technology family boundary




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# ASP.NET technology family boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  <TargetFramework>net10.0</TargetFramework>
2  <Nullable>enable</Nullable>
```

1 **Classic ASP**
Server-side script; legacy maintenance only

2 **Web Forms**
.aspx event lifecycle; not part of ASP.NET Core

3 **MVC 5**
System.Web on .NET Framework

4 **ASP.NET Core MVC**
Microsoft.AspNetCore.* packages

K1 / A1

Full procedures: Learner Guide


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TOPIC 1 · DESIGN DETAIL

# ASP.NET technology family boundary · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| Classic ASP | Server-side script; legacy maintenance only | Where is this enforced? |
| Web Forms | .aspx event lifecycle; not part of ASP.NET Core | What is observable? |
| MVC 5 | System.Web on .NET Framework | Who owns the value? |
| ASP.NET Core MVC | Microsoft.AspNetCore.* packages | What rejects invalid input? |
| Target | net10.0 for this course | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# ASP.NET technology family boundary · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Wrong tutorial** | System.Web examples do not compile on Core |
| **Wrong file** | .aspx controls are not Razor |
| **Wrong runtime** | .NET Framework is Windows-only |
| **Wrong host** | IIS concepts are mistaken for the app pipeline |

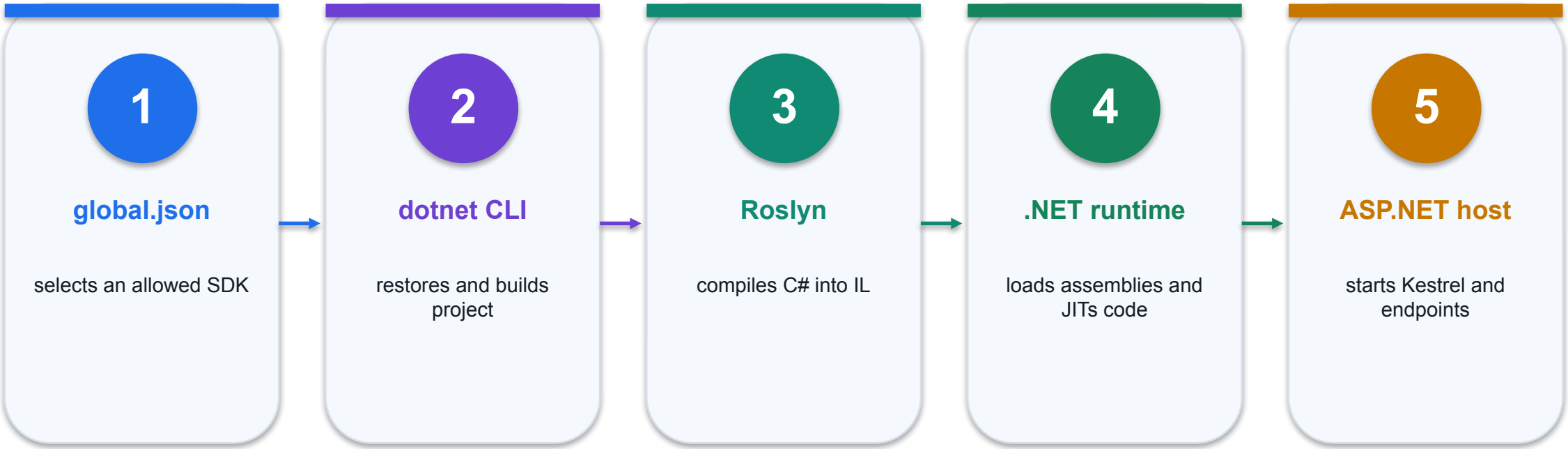
## ACCEPTANCE EVIDENCE

- ✓ dotnet --list-sdks includes 10.0
- ✓ csproj targets net10.0
- ✓ No System.Web dependency exists

**Record URL · command · status · build SHA**


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# SDK, runtime and target framework




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# SDK, runtime and target framework · worked artifact

REVIEWED EXCERPT · .NET 10

```
{
  "sdk": { "version": "10.0.100", "rollForward": "latestPatch" }
}
```

1. **SDK** — build tools plus templates
2. **Runtime** — executes a built application
3. **TFM** — net10.0 API compatibility contract
4. **RID** — optional platform-specific publish target

K1 / A1

Full procedures: Learner Guide


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# SDK, runtime and target framework · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| SDK | build tools plus templates | Where is this enforced? |
| Runtime | executes a built application | What is observable? |
| TFM | net10.0 API compatibility contract | Who owns the value? |
| RID | optional platform-specific publish target | What rejects invalid input? |
| Lock | global.json prevents silent SDK drift | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# SDK, runtime and target framework · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **NETSDK1045** | installed SDK cannot target net10.0 |
| **Restore error** | package source or version is unavailable |
| **Works on mine** | SDK version was not pinned |
| **Runtime missing** | framework-dependent publish reached an empty host |

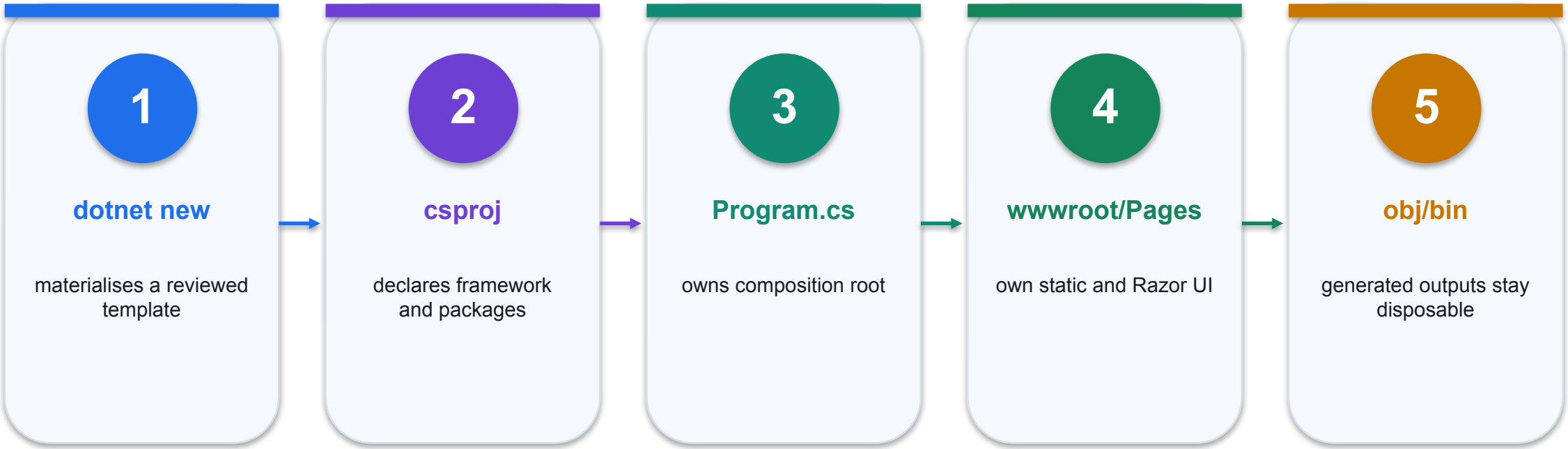
## ACCEPTANCE EVIDENCE

- ✓ dotnet --info records SDK and runtime
- ✓ dotnet build ends with 0 errors
- ✓ global.json is committed with the design

**Record URL · command · status · build SHA**


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# Project template and file ownership




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Project template and file ownership · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet new webapp -n LocalCRM --framework net10.0
2 dotnet new list web
```

1. **webapp**
   Razor Pages project template
2. **webapi**
   HTTP API template
3. **--no-https**
   only for constrained local tests
4. **--use-controllers**
   selects controller API style

K1 / A1

Full procedures: Learner Guide


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TOPIC 1 · DESIGN DETAIL

# Project template and file ownership · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| webapp | Razor Pages project template | Where is this enforced? |
| webapi | HTTP API template | What is observable? |
| --no-https | only for constrained local tests | Who owns the value? |
| --use-controllers | selects controller API style | What rejects invalid input? |
| --dry-run | previews files before creation | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 1 · DIAGNOSE + VERIFY

# Project template and file ownership · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Template absent** | required SDK/workload is missing |
| **Generated files edited** | obj/bin were mistaken for source |
| **Secret committed** | local settings were treated as portable |
| **Name collision** | namespace and folder identity diverged |

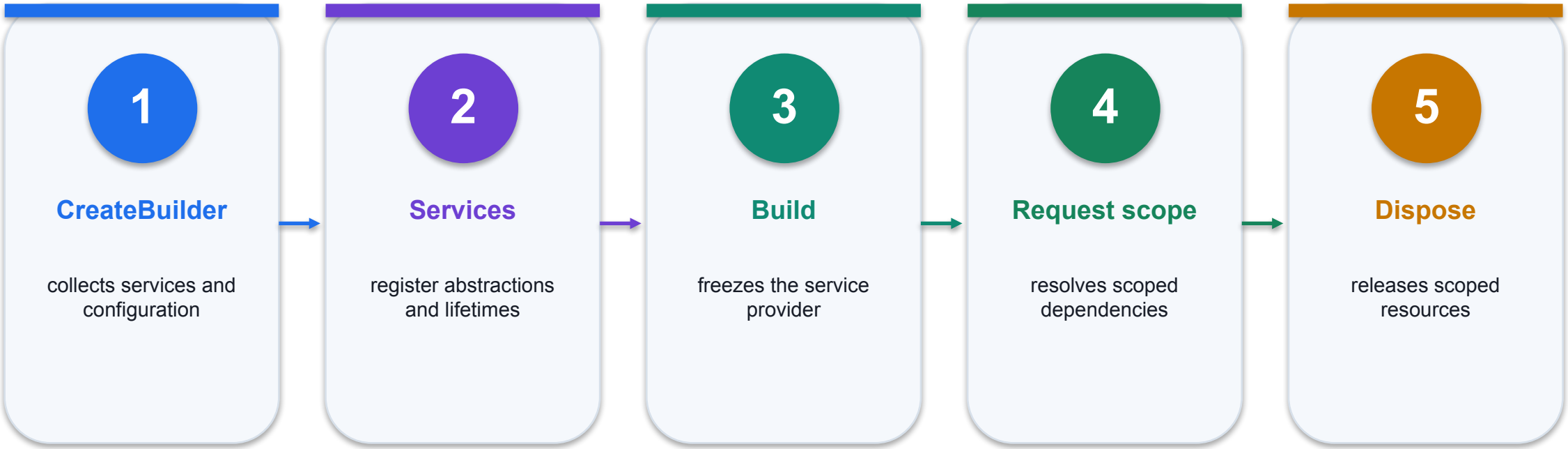
## ACCEPTANCE EVIDENCE

- ✓ dotnet new --dry-run shows the intended template
- ✓ git status excludes bin/ and obj/
- ✓ dotnet run exposes the generated endpoint

**Record URL · command · status · build SHA**


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# Composition root and dependency injection




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Composition root and dependency injection · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddScoped<IContactStore,
   SqliteContactStore>();
2  var app = builder.Build();
```

1. **Singleton**
   one instance for process lifetime
2. **Scoped**
   one instance per HTTP request
3. **Transient**
   new instance per resolution
4. **Constructor**
   declares required dependency

K1 / A1

Full procedures: Learner Guide


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TOPIC 1 · DESIGN DETAIL

# Composition root and dependency injection · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Singleton** | one instance for process lifetime | Where is this enforced? |
| **Scoped** | one instance per HTTP request | What is observable? |
| **Transient** | new instance per resolution | Who owns the value? |
| **Constructor** | declares required dependency | What rejects invalid input? |
| **Interface** | separates consumer from implementation | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Composition root and dependency injection · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Cannot resolve** | service was never registered |
| **Captive dependency** | singleton holds a scoped service |
| **Service locator** | runtime GetService hides the contract |
| **Duplicate registration** | last registration silently wins |

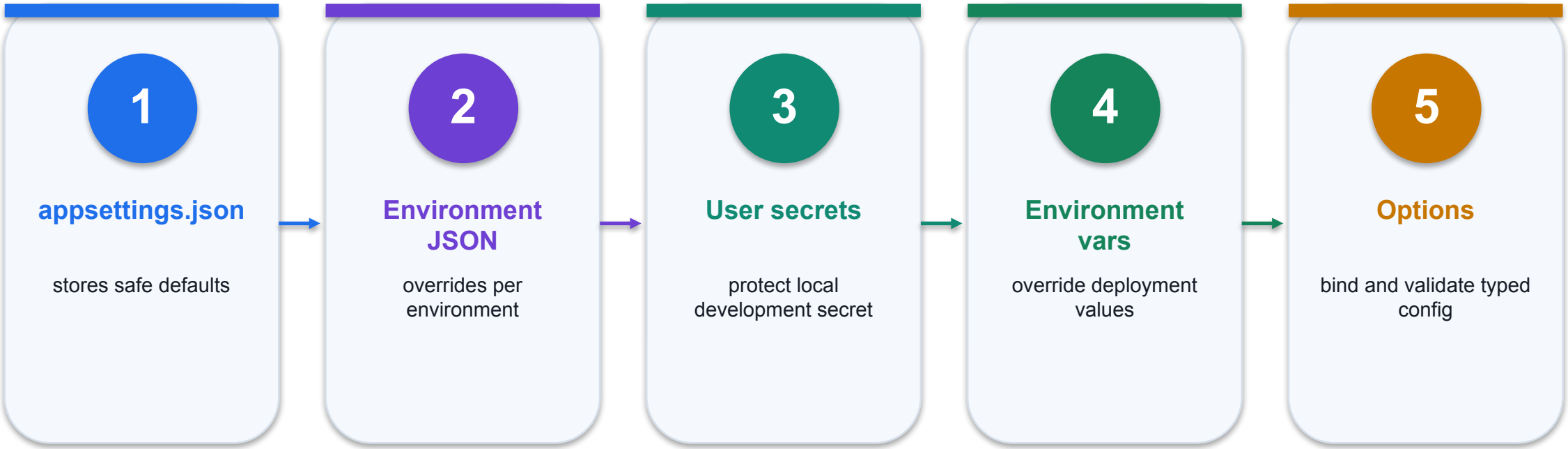
## ACCEPTANCE EVIDENCE

- ✓ Startup completes without DI exception
- ✓ One scoped instance is reused within a request
- ✓ Unit test can replace IContactStore

**Record URL · command · status · build SHA**


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# Configuration provider precedence




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Configuration provider precedence · worked artifact

REVIEWED EXCERPT · .NET 10

```
builder.Services.AddOptions<AppOptions>()
  .BindConfiguration("App")
  .ValidateDataAnnotations().ValidateOnStart();
```

1 **Key path**
ConnectionStrings:LocalCRM

2 **Env encoding**
double underscore represents colon

3 **Secret**
never committed to appsettings.json

4 **Options**
ValidateOnStart catches invalid config

K1 / A1

Full procedures: Learner Guide


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# Configuration provider precedence · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| Key path | ConnectionStrings:LocalCRM | Where is this enforced? |
| Env encoding | double underscore represents colon | What is observable? |
| Secret | never committed to appsettings.json | Who owns the value? |
| Options | ValidateOnStart catches invalid config | What rejects invalid input? |
| Precedence | later provider overrides earlier | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Configuration provider precedence · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Null option** | key path does not match property |
| **Dev leak** | secret landed in source control |
| **Wrong environment** | ASPNETCORE_ENVIRONMENT is incorrect |
| **Late failure** | options were not validated at startup |

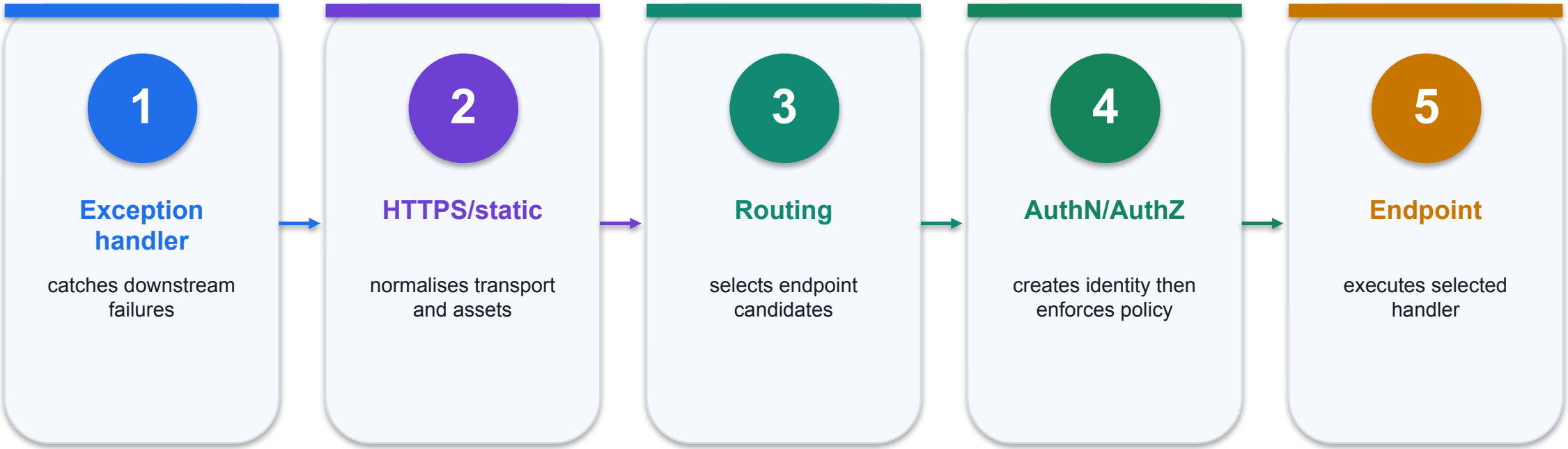
## ACCEPTANCE EVIDENCE

- ✓ Startup rejects an invalid required setting
- ✓ env var changes the resolved value
- ✓ secret scan finds no credential

**Record URL · command · status · build SHA**


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# Middleware pipeline ordering




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Middleware pipeline ordering · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 app.UseExceptionHandler("/Error");
2 app.UseHttpsRedirection();
3 app.UseAuthorization();
4 app.MapRazorPages();
```

1 **Use**
can run before and after next

2 **Run**
terminal middleware

3 **Map**
branches by path

4 **Order**
registration order is execution order

K1 / A1

Full procedures: Learner Guide


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TOPIC 1 · DESIGN DETAIL

# Middleware pipeline ordering · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Use** | can run before and after next | Where is this enforced? |
| **Run** | terminal middleware | What is observable? |
| **Map** | branches by path | Who owns the value? |
| **Order** | registration order is execution order | What rejects invalid input? |
| **Short-circuit** | response ends without downstream call | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 1 · DIAGNOSE + VERIFY

# Middleware pipeline ordering · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **401/403 surprise** | authorization ran before authentication |
| **Headers absent** | middleware ran after endpoint |
| **Static 404** | static files were not mapped |
| **Details leak** | developer exception page reached production |

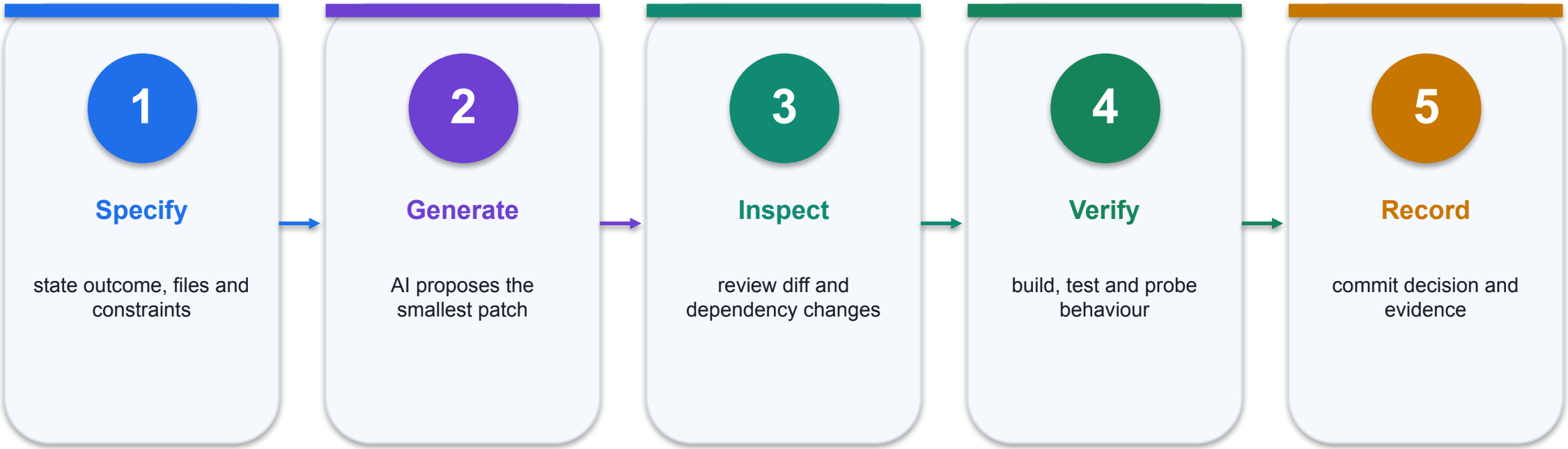
## ACCEPTANCE EVIDENCE

- ✓ Trace log shows expected middleware order
- ✓ Production errors use generic ProblemDetails
- ✓ Protected endpoint enforces its policy

**Record URL · command · status · build SHA**


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# Vibe coding control loop




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Vibe coding control loop · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 Prompt: Add GET /health to Program.cs only.
2 Constraints: net10.0; no new packages.
3 Verify: dotnet test && curl -i /health
```

1 **Context**
target files and accepted architecture

2 **Constraints**
net10.0, no secrets, accessible UI

3 **Acceptance**
commands and exact observable result

4 **Diff budget**
reject unrelated changes

K1 / A1

Full procedures: Learner Guide


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# Vibe coding control loop · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Context** | target files and accepted architecture | Where is this enforced? |
| **Constraints** | net10.0, no secrets, accessible UI | What is observable? |
| **Acceptance** | commands and exact observable result | Who owns the value? |
| **Diff budget** | reject unrelated changes | What rejects invalid input? |
| **Rollback** | checkpoint before generation | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Vibe coding control loop · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Prompt drift** | scope did not name files |
| **Hallucinated API** | package/version was not checked |
| **Silent regression** | only the happy path was probed |
| **Mega-diff** | one prompt mixed unrelated changes |

## ACCEPTANCE EVIDENCE

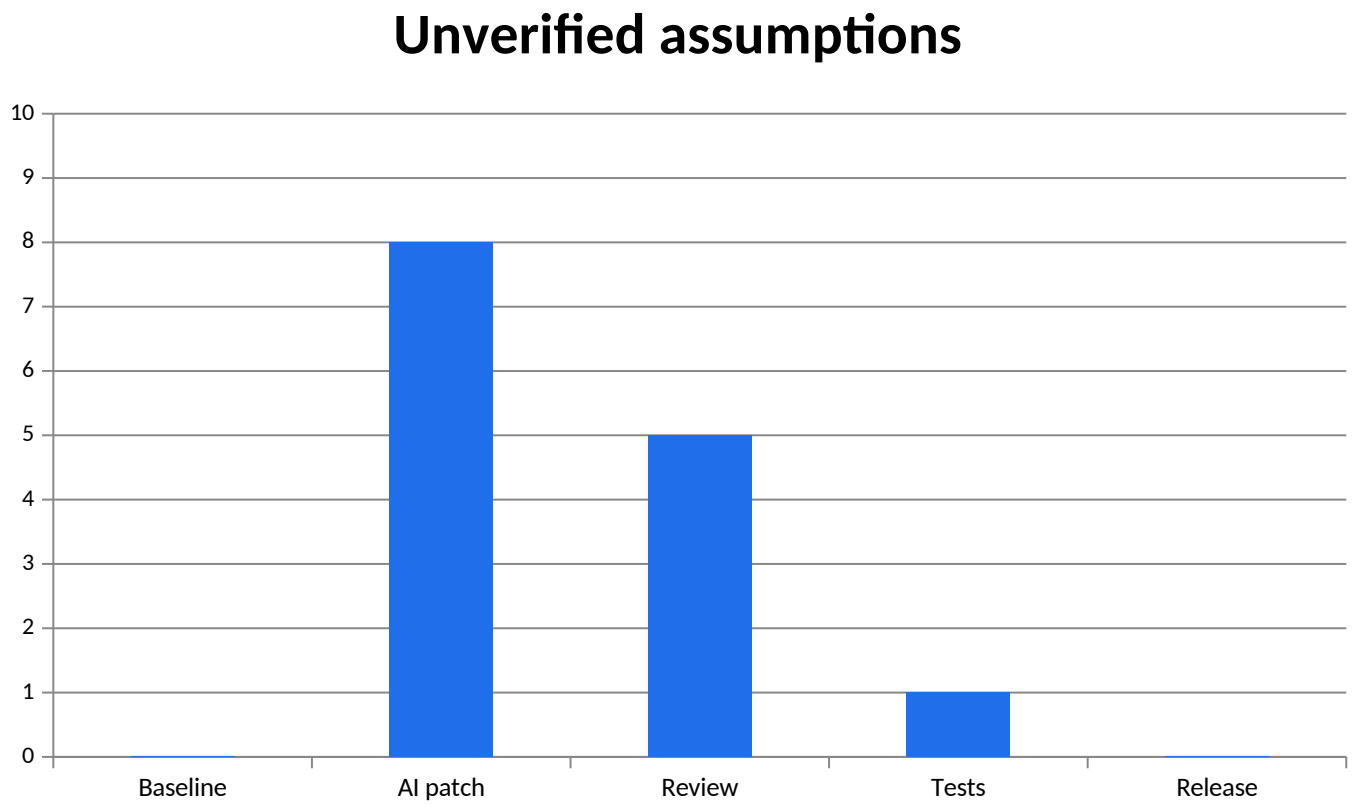
- ✓ git diff contains only declared files
- ✓ dotnet test is green
- ✓ manual probe produces named evidence

**Record URL · command · status · build SHA**


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TOPIC 1 · EDITABLE NATIVE CHART

# LO1 evidence profile




WHAT THE DATA SHOWS

**The review loop drives assumptions to zero before release.**

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

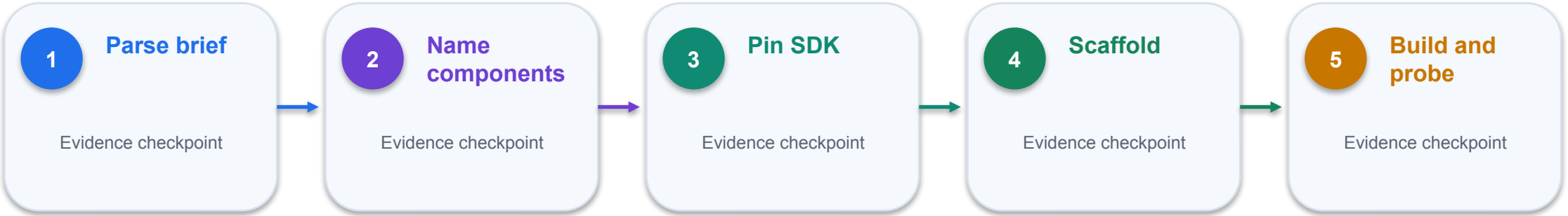
EDITABLE CHART


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55 MIN

# Activity 1: Translate Requirements into a Verified ASP.NET Core Scaffold

Turn a LocalCRM customer-contact brief into component boundaries and a reproducible .NET 10 Razor Pages project.




**YOU'LL PRODUCE** **requirements.md, component-map.md, pinned SDK file and a clean net10.0 Razor Pages scaffold**

Local snapshot: activities/activity-01-requirements-and-scaffold/src/LocalCRM · full commands in the Learner Guide


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# Activity 1 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Act as a software-design assistant. Convert the supplied LocalCRM brief into UI, application-service, data and hosting components. Preserve requirement IDs. Do not write code yet. Return assumptions separately.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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TOPIC 1 · VERIFY

# Activity 1 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: requirements.md, component-map.md, pinned SDK file and a clean net10.0 Razor Pages scaffold.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

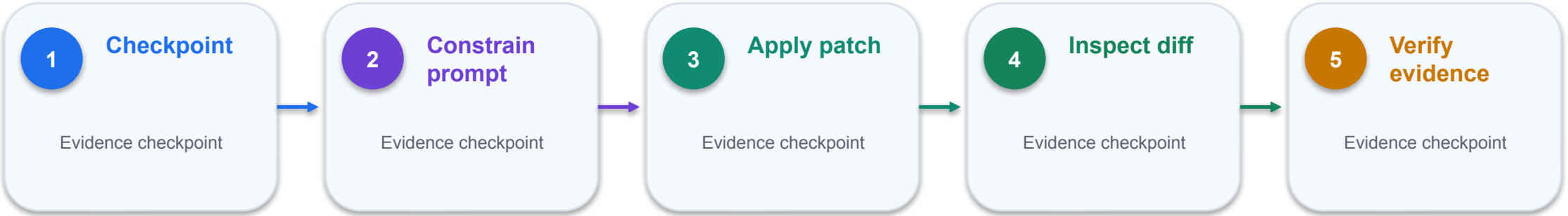
**K1/A1 EVIDENCE**


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45 MIN

# Activity 2: Run a Controlled Vibe-Coding Change

Add a health endpoint through a specify–generate–inspect–verify loop without accepting unrelated AI edits.




**YOU'LL PRODUCE** **a reviewed /health endpoint, build evidence and a decision log**

Local snapshot: activities/activity-02-controlled-vibe-change/src/LocalCRM · full commands in the Learner Guide


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# Activity 2 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Modify LocalCRM/Program.cs only. Add GET /health returning HTTP 200 JSON { status: 'ok', service: 'LocalCRM' }. Do not add packages or reorder existing middleware. Explain the exact diff and provide curl acceptance evidence.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 2 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: a reviewed /health endpoint, build evidence and a decision log.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K1/A1 EVIDENCE**


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LO1 · K1 / A1

# Topic 1 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.


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TOPIC 2 · LO2 · K2 / A2

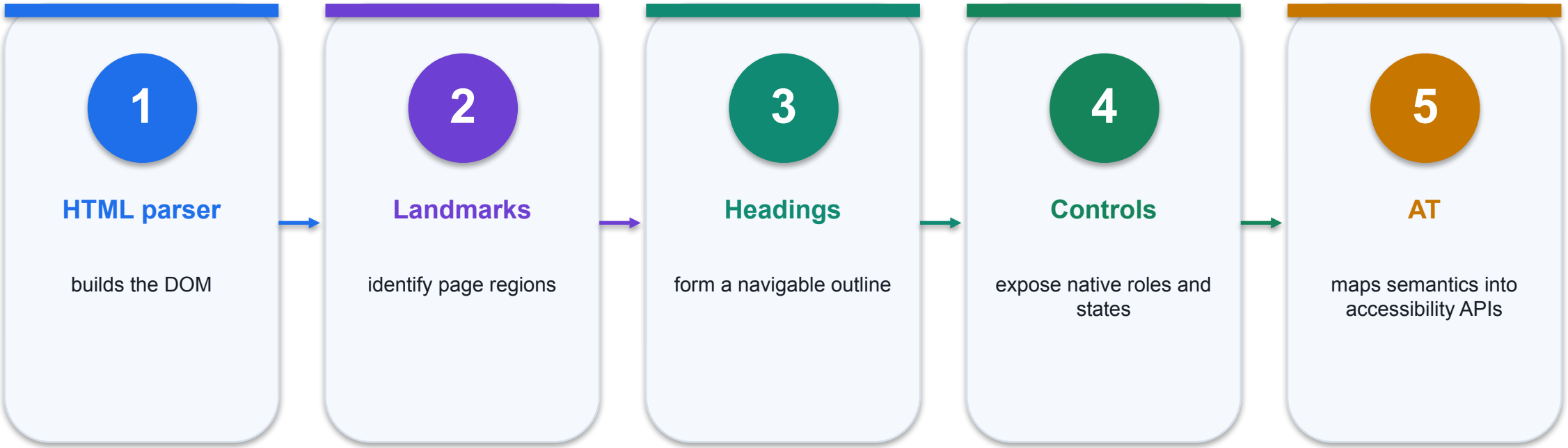
02

# Accessible UI and Organisational Vibe-Coding Practice

semantic structure → accessible interaction → reviewable AI-assisted change


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# Semantic document structure




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Semantic document structure · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <main>
2   <h1>CRM contacts</h1>
3   <button type="submit">Add contact</button>
4 </main>
```

1 **main**
one primary content region

2 **nav**
named navigation collection

3 **h1–h6**
do not skip levels for styling

4 **button**
action with keyboard behaviour

K2 / A2

Full procedures: Learner Guide


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# Semantic document structure · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| main | one primary content region | Where is this enforced? |
| nav | named navigation collection | What is observable? |
| h1–h6 | do not skip levels for styling | Who owns the value? |
| button | action with keyboard behaviour | What rejects invalid input? |
| a[href] | navigation to a real target | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Semantic document structure · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Div soup** | visual boxes have no programmatic role |
| **Heading jump** | outline no longer communicates hierarchy |
| **Clickable span** | keyboard activation is missing |
| **Duplicate main** | landmark navigation becomes ambiguous |

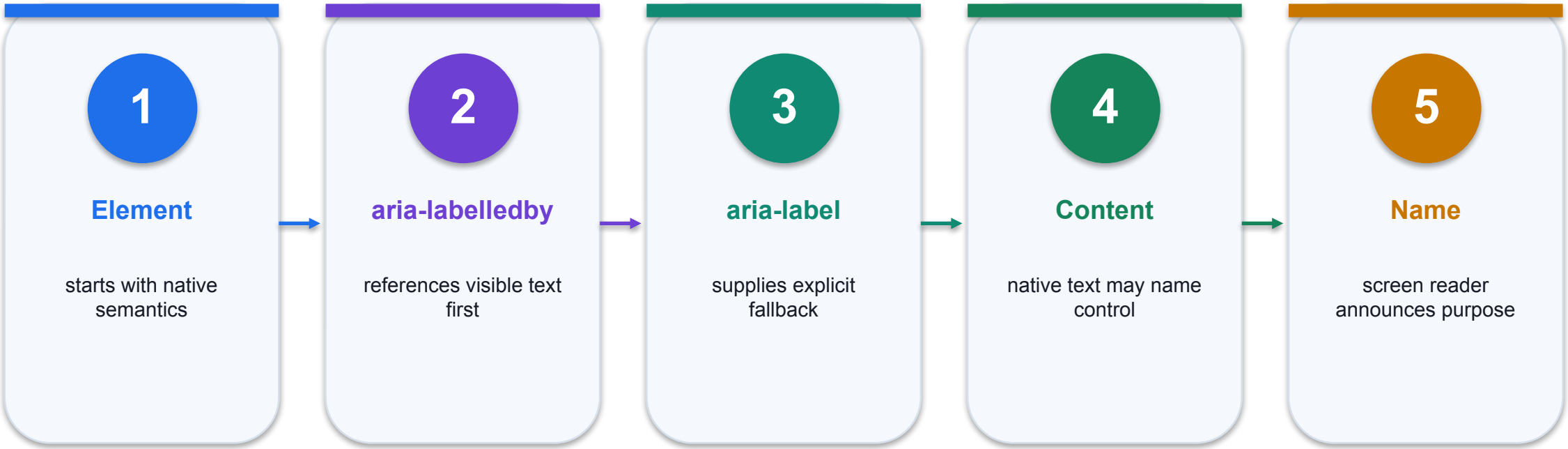
## ACCEPTANCE EVIDENCE

- ✓ Accessibility tree exposes main and button
- ✓ Heading outline has one h1
- ✓ Tab and Enter activate the control

**Record URL · command · status · build SHA**


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# Accessible name computation




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Accessible name computation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <label for="email">Summary</label>
2 <input id="email" name="email" type="email" aria-
  describedby="email-help">
```

1. **label for**
   associates text with an input id
2. **alt**
   names an informative image
3. **empty alt**
   marks decorative image
4. **aria-describedby**
   adds help, not the name

K2 / A2

Full procedures: Learner Guide


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# Accessible name computation · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| label for | associates text with an input id | Where is this enforced? |
| alt | names an informative image | What is observable? |
| empty alt | marks decorative image | Who owns the value? |
| aria-describedby | adds help, not the name | What rejects invalid input? |
| Unique id | prevents broken reference | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Accessible name computation · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Placeholder only** | name disappears after typing |
| **Filename announced** | image has missing or poor alt |
| **Duplicate id** | label binds to the wrong control |
| **ARIA override** | visible and spoken names disagree |

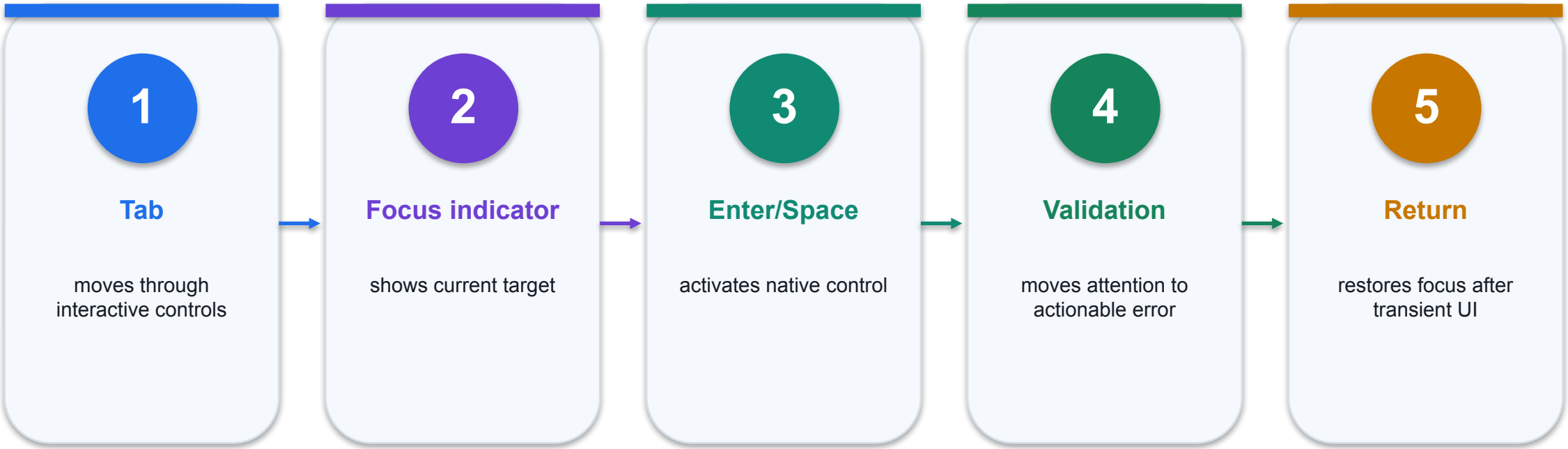
## ACCEPTANCE EVIDENCE

- ✓ Browser Accessibility pane shows the intended name
- ✓ Screen reader announces label and help
- ✓ axe reports no form-label violation

**Record URL · command · status · build SHA**


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# Keyboard and focus contract




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Keyboard and focus contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  button:focus-visible { outline: 3px solid #1f6feb; outline-
   offset: 3px; }
```

1 **Order**
DOM order should match visual order

2 **tabindex=0**
adds custom widget to natural order

3 **tabindex=-1**
allows scripted focus only

4 **:focus-visible**
keeps visible keyboard cue

K2 / A2

Full procedures: Learner Guide


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# Keyboard and focus contract · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| Order | DOM order should match visual order | Where is this enforced? |
| tabindex=0 | adds custom widget to natural order | What is observable? |
| tabindex=-1 | allows scripted focus only | Who owns the value? |
| :focus-visible | keeps visible keyboard cue | What rejects invalid input? |
| Escape | closes modal/menu and restores focus | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Keyboard and focus contract · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Focus trap** | user cannot leave a component |
| **Focus loss** | removed node held focus |
| **Positive tabindex** | manual ordering becomes brittle |
| **Mouse-only handler** | keyboard event path is absent |

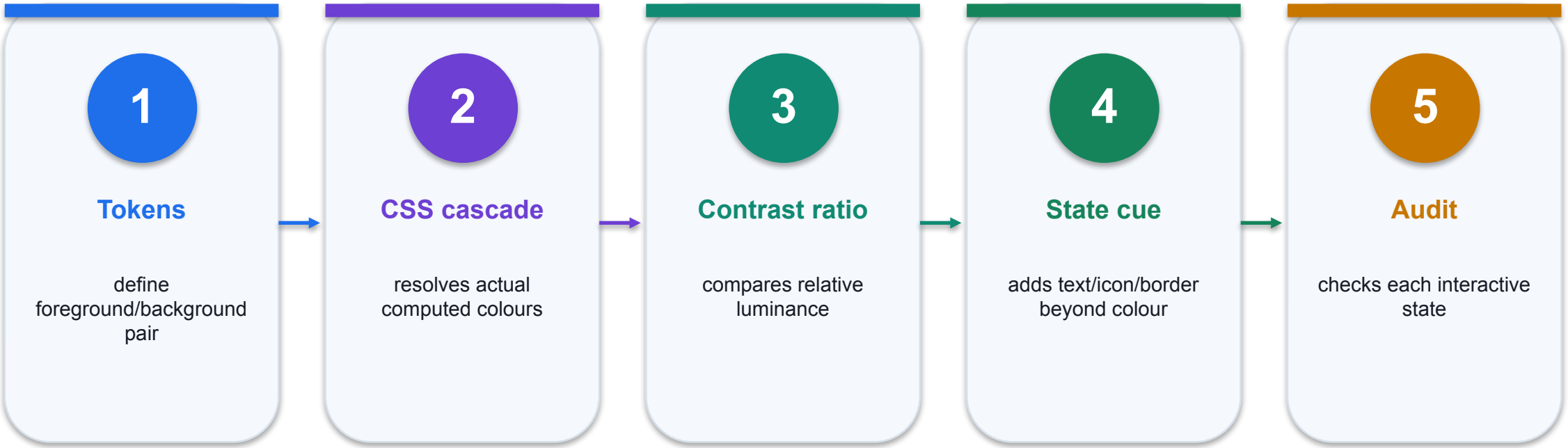
## ACCEPTANCE EVIDENCE

- ✓ Entire workflow completes without mouse
- ✓ Focus indicator remains visible
- ✓ Modal Escape returns focus to opener

**Record URL · command · status · build SHA**


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# Contrast and non-colour cues




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Contrast and non-colour cues · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  .error { color:#b42318; border-left:4px solid #b42318; }
2  .error::before { content:"Error: "; font-weight:700; }
```

1 **Normal text**
target 4.5:1

2 **Large text**
target 3:1

3 **UI component**
target 3:1 against adjacent colour

4 **Link**
not identified by colour alone

K2 / A2

Full procedures: Learner Guide


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# Contrast and non-colour cues · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Normal text** | target 4.5:1 | Where is this enforced? |
| **Large text** | target 3:1 | What is observable? |
| **UI component** | target 3:1 against adjacent colour | Who owns the value? |
| **Link** | not identified by colour alone | What rejects invalid input? |
| **State** | hover, focus, disabled and error all checked | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Contrast and non-colour cues · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Token passes only once** | hover state drops contrast |
| **Red/green only** | colour vision users lose status |
| **Text on image** | background changes under glyphs |
| **Disabled too faint** | essential text becomes unreadable |

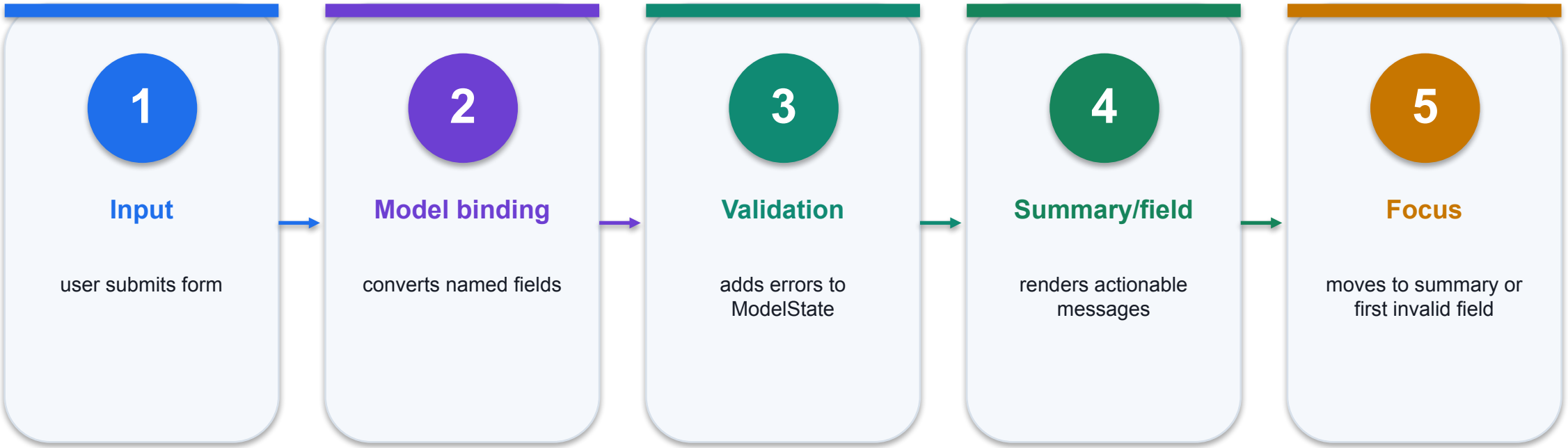
## ACCEPTANCE EVIDENCE

- ✓ Contrast checker records each token pair
- ✓ Status remains understandable in grayscale
- ✓ Lighthouse/axe result is attached

**Record URL · command · status · build SHA**


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# Validation error experience




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Validation error experience · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <span asp-validation-for="Input.Email"></span>
2 <div asp-validation-summary="ModelOnly" tabindex="-1"></div>
```

1. **asp-validation-for**
   field-specific message
2. **asp-validation-summary**
   page-level error collection
3. **aria-invalid**
   marks failing input
4. **describedby**
   associates message

K2 / A2

Full procedures: Learner Guide


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# Validation error experience · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| asp-validation-for | field-specific message | Where is this enforced? |
| asp-validation-summary | page-level error collection | What is observable? |
| aria-invalid | marks failing input | Who owns the value? |
| describedby | associates message | What rejects invalid input? |
| Server check | remains authoritative without JavaScript | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Validation error experience · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Client-only** | bypassed request accepts invalid data |
| **Generic error** | user cannot identify repair |
| **Focus unchanged** | screen-reader user misses result |
| **Values lost** | failed post rebuilds an empty form |

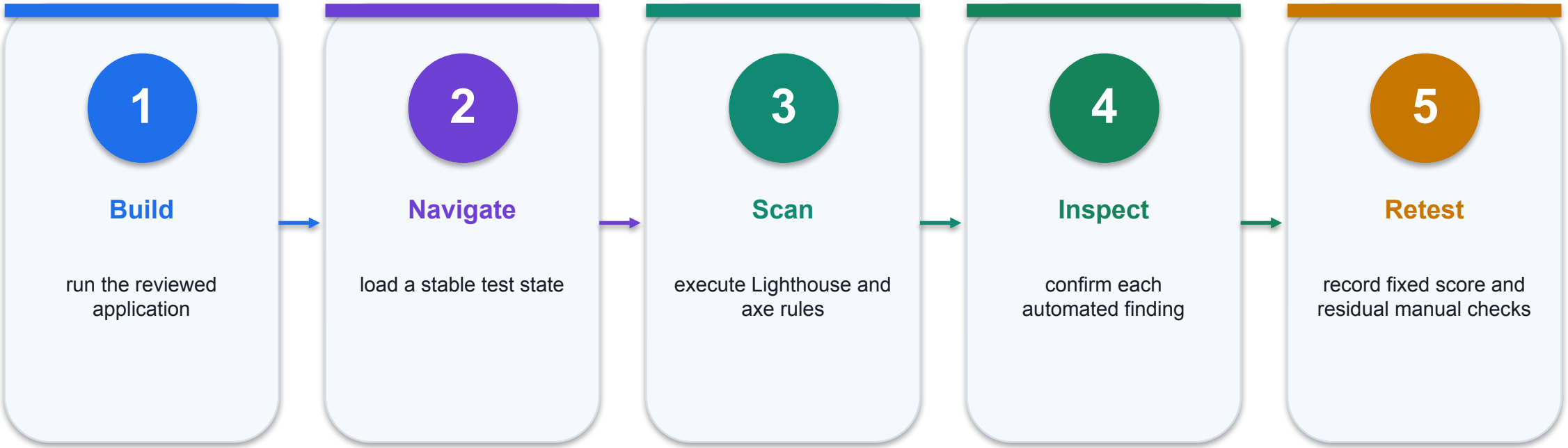
## ACCEPTANCE EVIDENCE

- ✓ Empty submit shows specific message
- ✓ Error is associated with the field
- ✓ Server rejects the same invalid payload

**Record URL · command · status · build SHA**


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# Lighthouse and axe evidence workflow




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Lighthouse and axe evidence workflow · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 npx @axe-core/cli http://localhost:5000 --exit
2 # Expected: 0 violations (manual checks still required)
```

1 **Automated**
finds detectable rule violations

2 **Manual**
keyboard, reading order and meaning

3 **Baseline**
store score and violation IDs

4 **Regression**
fail build on new serious violations

K2 / A2

Full procedures: Learner Guide


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# Lighthouse and axe evidence workflow · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Automated** | finds detectable rule violations | Where is this enforced? |
| **Manual** | keyboard, reading order and meaning | What is observable? |
| **Baseline** | store score and violation IDs | Who owns the value? |
| **Regression** | fail build on new serious violations | What rejects invalid input? |
| **Evidence** | HTML/JSON report plus screenshot | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Lighthouse and axe evidence workflow · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **False confidence** | score was treated as full conformance |
| **Flaky state** | scan ran before page settled |
| **Third-party noise** | scope was not documented |
| **No baseline** | regression cannot be distinguished |

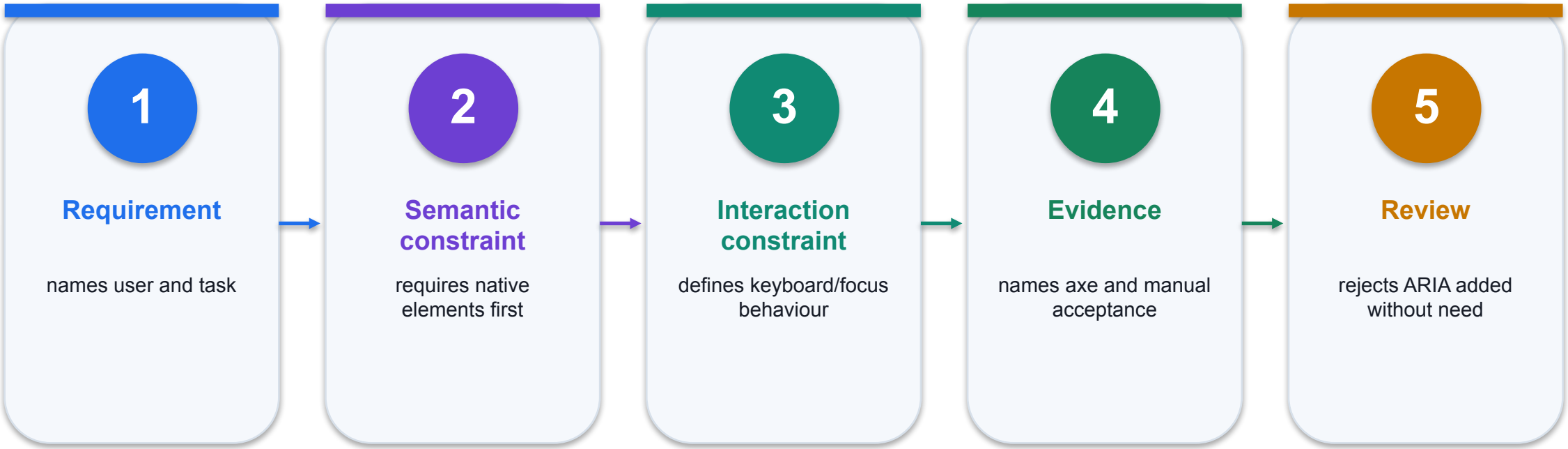
## ACCEPTANCE EVIDENCE

- ✓ axe exits 0 for the target state
- ✓ Manual keyboard checklist is signed
- ✓ Report names URL, date and build SHA

**Record URL · command · status · build SHA**


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# Prompt constraints for accessible output




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Prompt constraints for accessible output · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 Prompt: Make Pages/Contacts/Create.cshtml keyboard complete.
2 Use native controls; preserve label text.
3 Acceptance: axe=0 and Tab→Enter submits.
```

1 **Files**
limit AI to the intended Razor/CSS files

2 **Standard**
reference WCAG outcome, not vague accessibility

3 **Prohibition**
no clickable divs or positive tabindex

4 **Acceptance**
keyboard path plus automated scan

K2 / A2

Full procedures: Learner Guide


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# Prompt constraints for accessible output · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| **Files** | limit AI to the intended Razor/CSS files | Where is this enforced? |
| **Standard** | reference WCAG outcome, not vague accessibility | What is observable? |
| **Prohibition** | no clickable divs or positive tabindex | Who owns the value? |
| **Acceptance** | keyboard path plus automated scan | What rejects invalid input? |
| **Output** | request patch and rationale, not prose only | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Prompt constraints for accessible output · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Vague prompt** | model optimises appearance only |
| **ARIA cargo cult** | redundant roles conflict with native semantics |
| **Hidden focus** | CSS reset removes outline |
| **Untested claim** | AI says accessible without evidence |

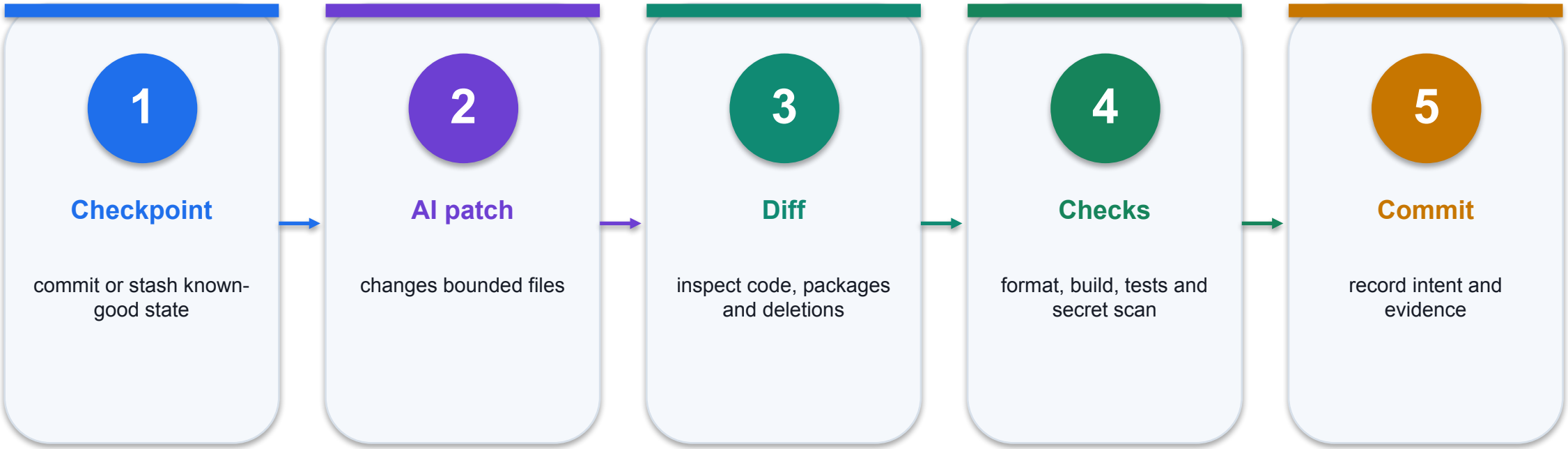
## ACCEPTANCE EVIDENCE

- ✓ Diff uses native elements
- ✓ Tab order matches DOM order
- ✓ Acceptance evidence is reproducible

**Record URL · command · status · build SHA**


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# Git diff and review guardrails




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Git diff and review guardrails · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 git diff --check
2 dotnet format --verify-no-changes
3 dotnet test
4 git diff --name-only
```

1. **git diff --stat**
   detects unexpected breadth
2. **git diff --check**
   finds whitespace defects
3. **dotnet format --verify-no-changes**
   enforces style
4. **dotnet test**
   proves behavioural contract

K2 / A2

Full procedures: Learner Guide


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# Git diff and review guardrails · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| git diff --stat | detects unexpected breadth | Where is this enforced? |
| git diff --check | finds whitespace defects | What is observable? |
| dotnet format --verify-no-changes | enforces style | Who owns the value? |
| dotnet test | proves behavioural contract | What rejects invalid input? |
| Secret scan | blocks credentials before history | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Git diff and review guardrails · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Generated noise** | bin/obj entered the diff |
| **Package surprise** | AI added dependency without review |
| **Secret leak** | connection string was committed |
| **Mixed commit** | unrelated changes obscure review |

## ACCEPTANCE EVIDENCE

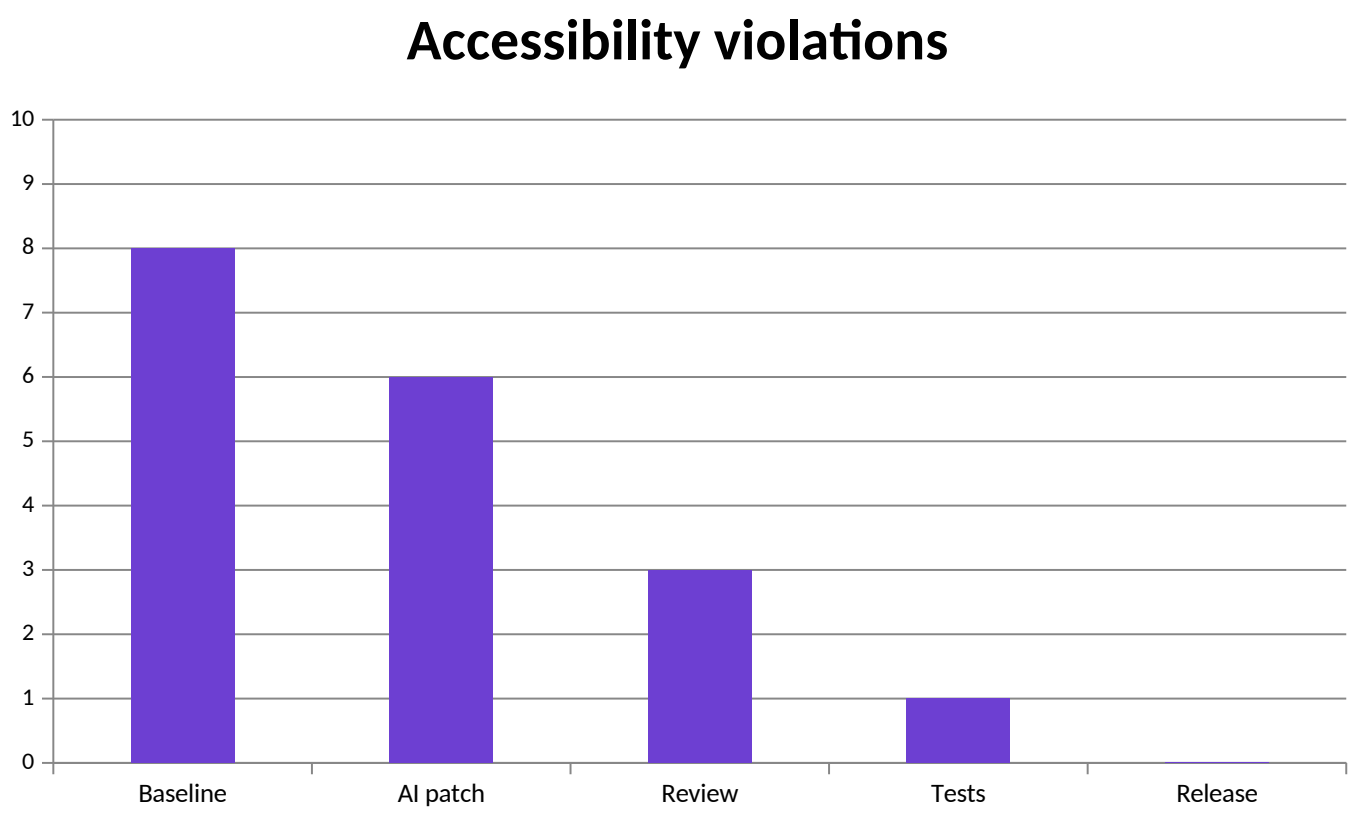
- ✓ Only scoped source files changed
- ✓ All local gates exit 0
- ✓ Commit message links requirement and evidence

**Record URL · command · status · build SHA**


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TOPIC 2 · EDITABLE NATIVE CHART

# LO2 evidence profile




WHAT THE DATA SHOWS

**Automated evidence improves; manual keyboard checks still remain mandatory.**

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

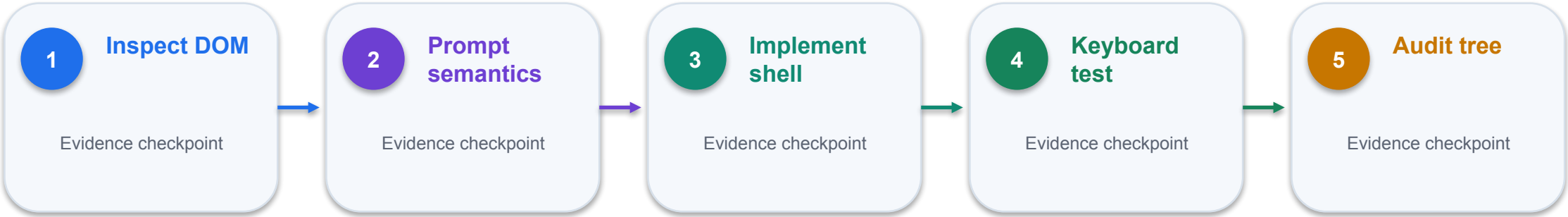
EDITABLE CHART


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55 MIN

# Activity 3: Build an Accessible Razor Pages Shell

Create semantic navigation, visible keyboard focus and a labelled customer-contact entry point.




**YOU'LL PRODUCE** **an accessible layout and Contacts index page with a reproducible keyboard checklist**

Local snapshot: activities/activity-03-accessible-razor-shell/src/LocalCRM · full commands in the Learner Guide


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# Activity 3 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Update the Razor layout and add Pages/Contacts/Index.cshtml. Use native header/nav/main/footer landmarks, one h1, descriptive link text and visible :focus-visible styles. No clickable divs, positive tabindex or ARIA when native HTML suffices.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 3 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: an accessible layout and Contacts index page with a reproducible keyboard checklist.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

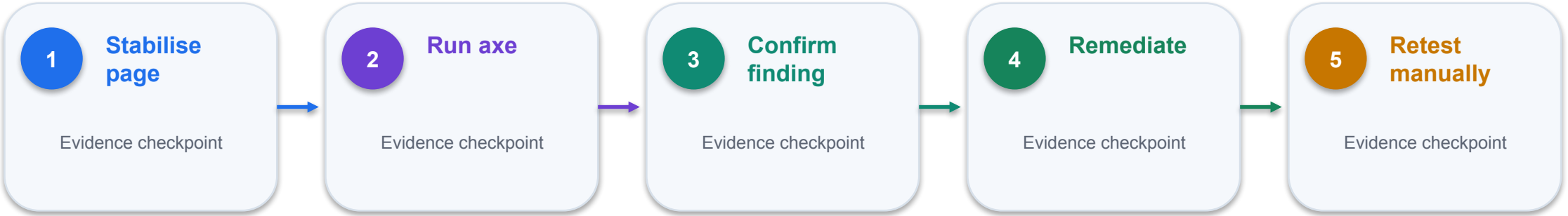
**K2/A2 EVIDENCE**


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45 MIN

# Activity 4: Audit Accessibility with axe and Manual Evidence

Create a repeatable automated-plus-manual accessibility gate for the running Razor Pages UI.




**YOU'LL PRODUCE** **axe JSON evidence, a manual keyboard checklist and a reviewed remediation diff**

Local snapshot: activities/activity-04-accessibility-audit/src/LocalCRM · full commands in the Learner Guide


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# Activity 4 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Review the supplied axe findings and the Razor markup. Propose the smallest semantic fix for each confirmed violation. Do not suppress rules. State which outcomes still require manual keyboard or screen-reader checks.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 4 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: axe JSON evidence, a manual keyboard checklist and a reviewed remediation diff.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K2/A2 EVIDENCE**


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LO2 · K2 / A2

# Topic 2 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.


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TOPIC 3 · LO3 · K3 / A3

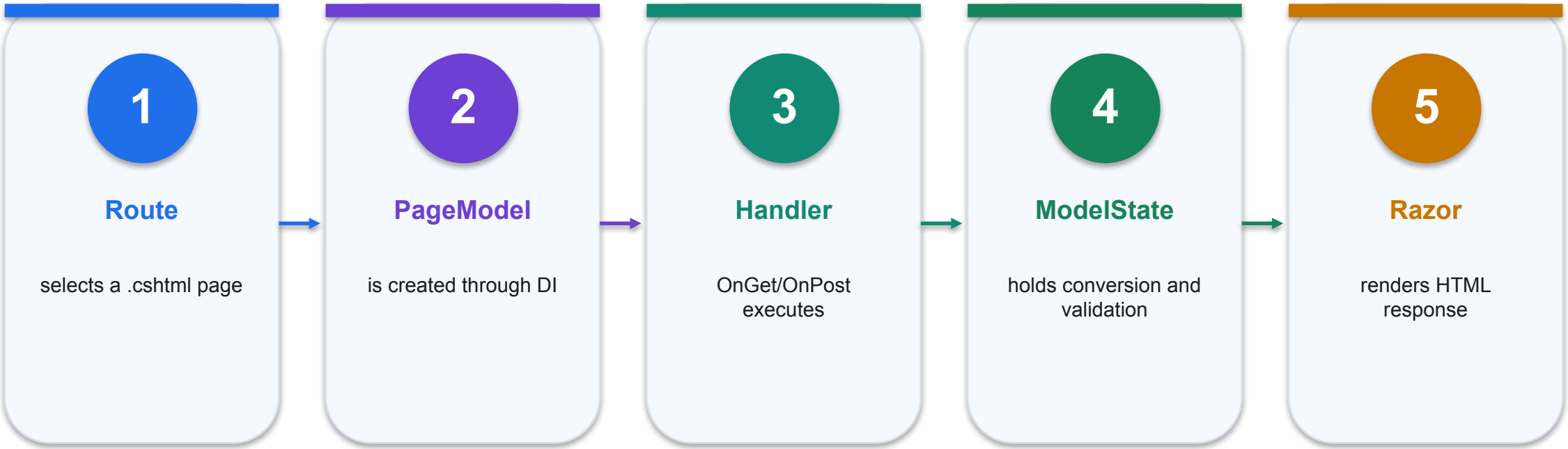
# Razor Pages Controls, Binding and Persistence

03

page-focused UI → binding and validation → antiforgery → durable state


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# Razor Page execution model




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Razor Page execution model · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 public class CreateModel(IContactStore store) : PageModel
2 { public void OnGet() { }
3   public async Task<IActionResult> OnPostAsync() {
/*...*/ } }
```

1 **.cshtml**
markup plus directives and Tag Helpers

2 **.cshtml.cs**
page handlers and input model

3 **OnGet**
safe read path

4 **OnPost**
form mutation path

K3 / A3

Full procedures: Learner Guide


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# Razor Page execution model · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| .cshtml | markup plus directives and Tag Helpers | Where is this enforced? |
| .cshtml.cs | page handlers and input model | What is observable? |
| OnGet | safe read path | Who owns the value? |
| OnPost | form mutation path | What rejects invalid input? |
| IActionResult | redirect, page or status result | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Razor Page execution model · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Handler not called** | method/name convention mismatched |
| **Null service** | DI registration missing |
| **Double submit** | post returned Page instead of PRG redirect |
| **Logic in view** | separation of concerns collapsed |

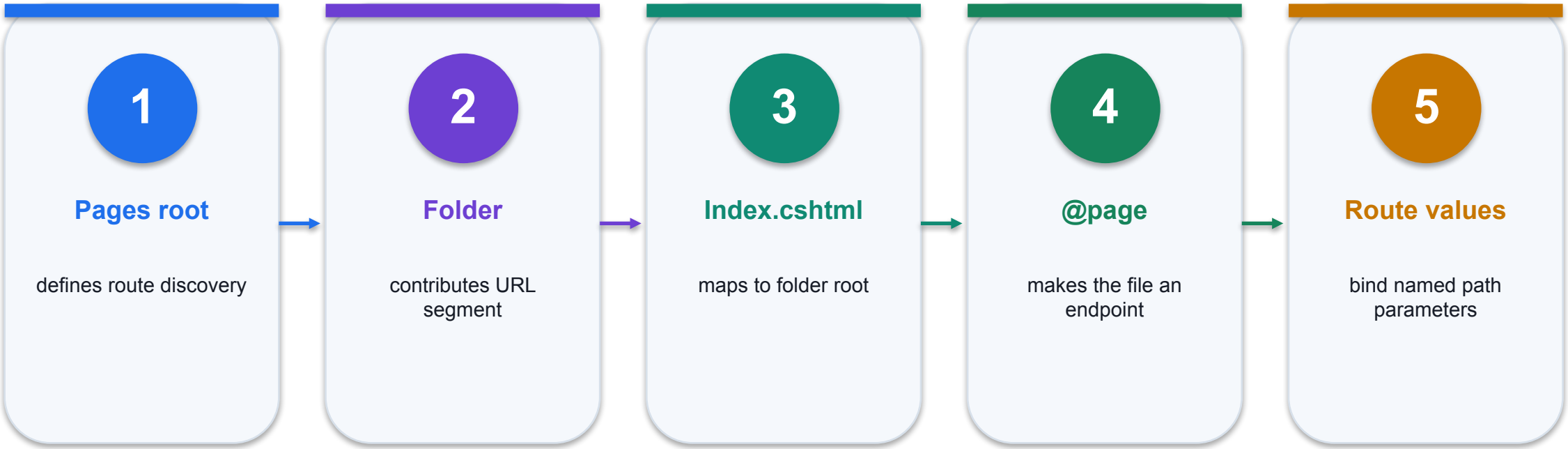
## ACCEPTANCE EVIDENCE

- ✓ GET selects OnGet
- ✓ POST selects OnPostAsync
- ✓ Successful POST redirects to index

**Record URL · command · status · build SHA**


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# Razor Pages routing conventions




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Razor Pages routing conventions · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 @page "{id:int}"
2 <a asp-page="./Details" asp-route-id="@item.Id">View</a>
```

1 **/Pages/Contacts/Index.cshtml**
route /Contacts

2 **@page "{id:int}"**
typed route constraint

3 **asp-page**
generates page URL

4 **asp-route-id**
adds route value

K3 / A3

Full procedures: Learner Guide


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# Razor Pages routing conventions · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| /Pages/Contacts/ Index.cshtml | route /Contacts | Where is this enforced? |
| @page "{id:int}" | typed route constraint | What is observable? |
| asp-page | generates page URL | Who owns the value? |
| asp-route-id | adds route value | What rejects invalid input? |
| Areas | partition larger applications | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 3 · DIAGNOSE + VERIFY

# Razor Pages routing conventions · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **404** | @page is missing or path is wrong |
| **Ambiguous link** | relative page path resolves elsewhere |
| **String id** | route constraint was omitted |
| **Hardcoded URL** | renaming page breaks navigation |

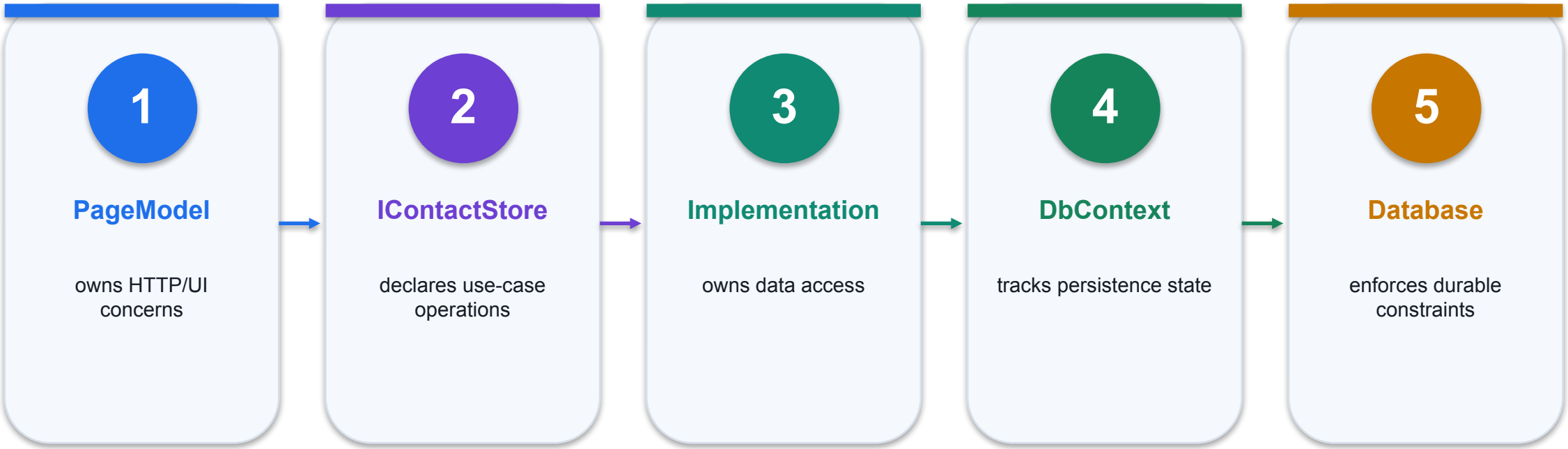
## ACCEPTANCE EVIDENCE

- ✓ LinkGenerator produces /Contacts/42
- ✓ Non-integer id does not match
- ✓ Renaming folder requires no hardcoded href edit

**Record URL · command · status · build SHA**


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# Service boundary behind the PageModel




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Service boundary behind the PageModel · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 public interface IContactStore
2 { Task<int> AddAsync(NewContact input, CancellationToken ct);
3   Task<IReadOnlyList<ContactRow>> ListAsync(CancellationToken
ct); }
```

**1 Interface**

Task<List<CustomerContact>> ListAsync

**2 Cancellation**

accept RequestAborted token

**3 DTO/ViewModel**

limits exposed fields

**4 Scoped lifetime**

one DbContext per request

K3 / A3

Full procedures: Learner Guide


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# Service boundary behind the PageModel · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| Interface | Task<List<CustomerContact>> ListAsync | Where is this enforced? |
| Cancellation | accept RequestAborted token | What is observable? |
| DTO/ViewModel | limits exposed fields | Who owns the value? |
| Scoped lifetime | one DbContext per request | What rejects invalid input? |
| No static state | supports concurrent requests | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Service boundary behind the PageModel · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Fat PageModel** | SQL and presentation logic are mixed |
| **Static list** | data races and restart loss |
| **Sync-over-async** | .Result can block request threads |
| **Leaky entity** | persistence shape becomes public contract |

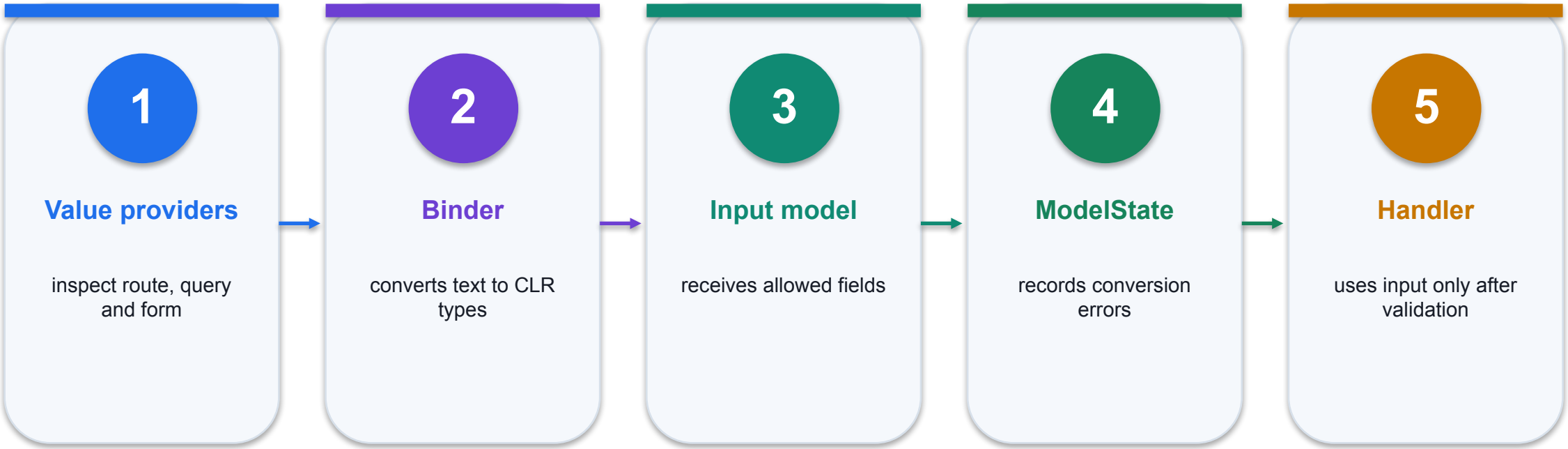
## ACCEPTANCE EVIDENCE

- ✓ PageModel unit test substitutes IContactStore
- ✓ Cancellation reaches EF Core
- ✓ Concurrent requests do not share mutable list

**Record URL · command · status · build SHA**


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# Model binding input sources




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Model binding input sources · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [BindProperty] public NewContact Input { get; set; } = new();
2 public sealed record NewContact(string Summary, string
  Category);
```

1 **[BindProperty]**
binds form property for non-GET by default

2 **SupportsGet**
explicitly allows GET property binding

3 **TryParse**
controls custom conversion

4 **Prefix**
matches nested field names

K3 / A3

Full procedures: Learner Guide


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# Model binding input sources · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| [BindProperty] | binds form property for non-GET by default | Where is this enforced? |
| SupportsGet | explicitly allows GET property binding | What is observable? |
| TryParse | controls custom conversion | Who owns the value? |
| Prefix | matches nested field names | What rejects invalid input? |
| Overposting | prevented by dedicated input model | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Model binding input sources · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Default values** | conversion error was ignored |
| **Overposting** | entity was bound directly |
| **Wrong prefix** | form name does not match property path |
| **GET mutation** | SupportsGet was misused for a command |

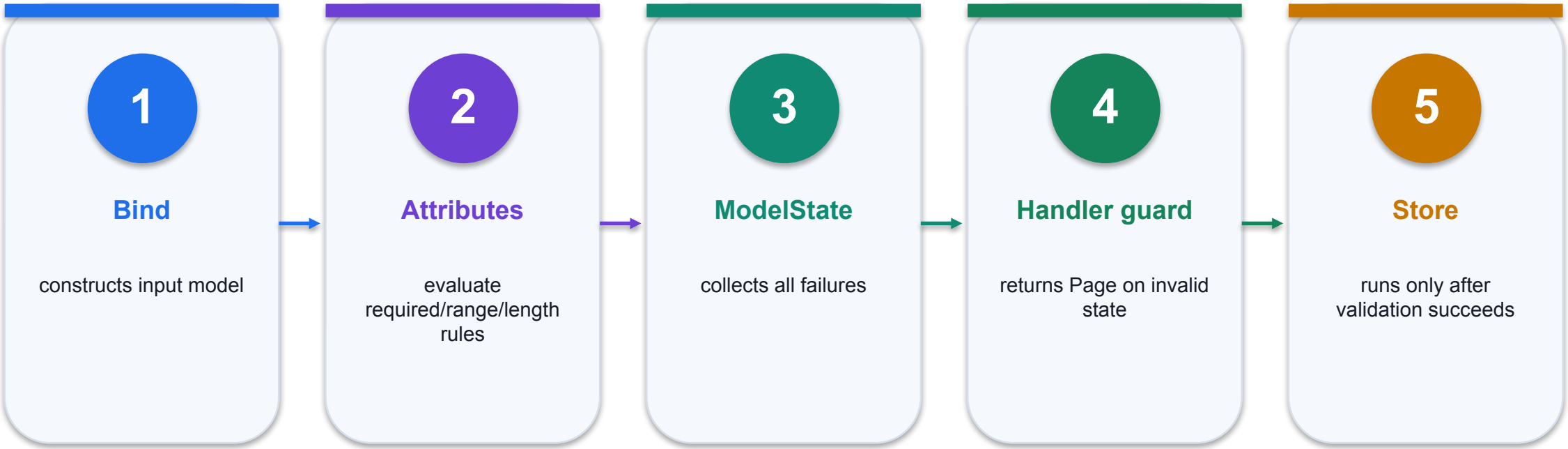
## ACCEPTANCE EVIDENCE

- ✓ Invalid integer creates ModelState error
- ✓ Unexpected form field is ignored
- ✓ Valid form populates only NewContact fields

**Record URL · command · status · build SHA**


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# DataAnnotations and server validation




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# DataAnnotations and server validation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [Required, EmailAddress]
2 public string Email { get; init; } = string.Empty;
```

1 **[Required]**
rejects missing/empty value

2 **[StringLength]**
sets bounded text contract

3 **[Range]**
sets numeric limits

4 **IValidatableObject**
supports cross-field rule

K3 / A3

Full procedures: Learner Guide


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# DataAnnotations and server validation · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| [Required] | rejects missing/empty value | Where is this enforced? |
| [StringLength] | sets bounded text contract | What is observable? |
| [Range] | sets numeric limits | Who owns the value? |
| IValidatableObject | supports cross-field rule | What rejects invalid input? |
| Database | still enforces storage constraints | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# DataAnnotations and server validation · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Client bypass** | server guard was omitted |
| **Mismatched length** | UI and database limits differ |
| **Opaque message** | default error lacks recovery guidance |
| **Duplicate rule** | business validation diverges across endpoints |

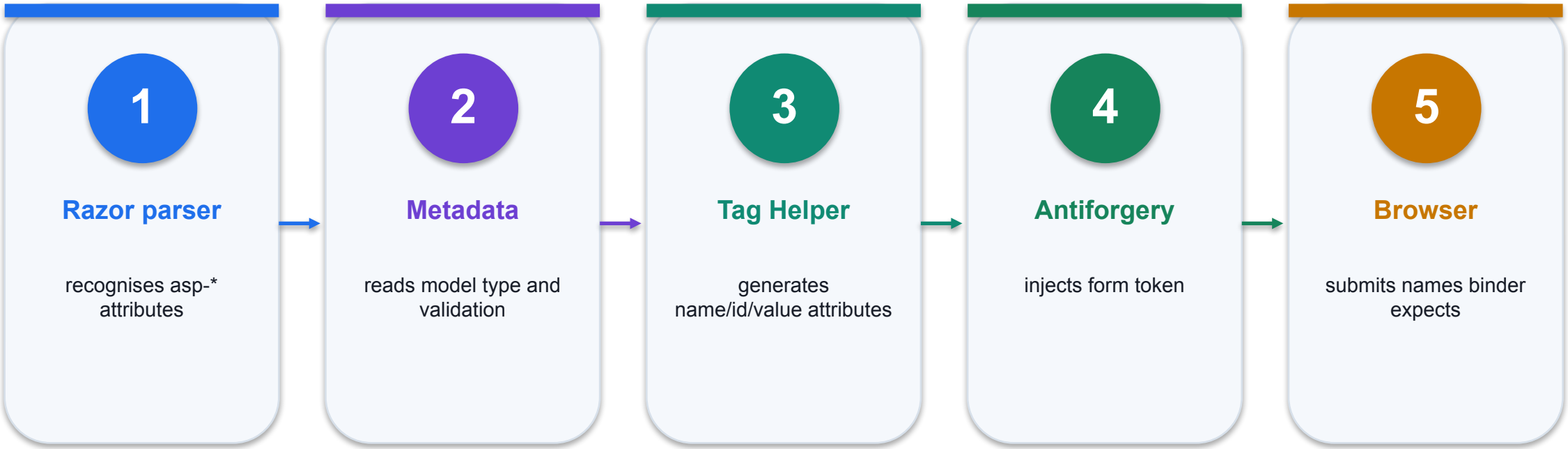
## ACCEPTANCE EVIDENCE

- ✓ 4-character summary is rejected
- ✓ 120-character summary succeeds
- ✓ Store mock receives no call for invalid input

**Record URL · command · status · build SHA**


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# Tag Helpers generate consistent controls




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Tag Helpers generate consistent controls · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <label asp-for="Input.Email"></label>
2 <input asp-for="Input.Email" />
3 <span asp-validation-for="Input.Email"></span>
```

1 **asp-for**
binds label/input/message to one expression

2 **asp-page**
generates route-aware URL

3 **asp-validation-for**
renders ModelState error

4 **form method=post**
participates in antiforgery

K3 / A3

Full procedures: Learner Guide


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# Tag Helpers generate consistent controls · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| asp-for | binds label/input/message to one expression | Where is this enforced? |
| asp-page | generates route-aware URL | What is observable? |
| asp-validation-for | renders ModelState error | Who owns the value? |
| form method=post | participates in antiforgery | What rejects invalid input? |
| partial | reuses reviewed markup | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Tag Helpers generate consistent controls · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Manual name drift** | binder receives no value |
| **Duplicate id** | labels and errors bind incorrectly |
| **Raw URL** | route refactor breaks link |
| **Missing partial** | validation scripts never load |

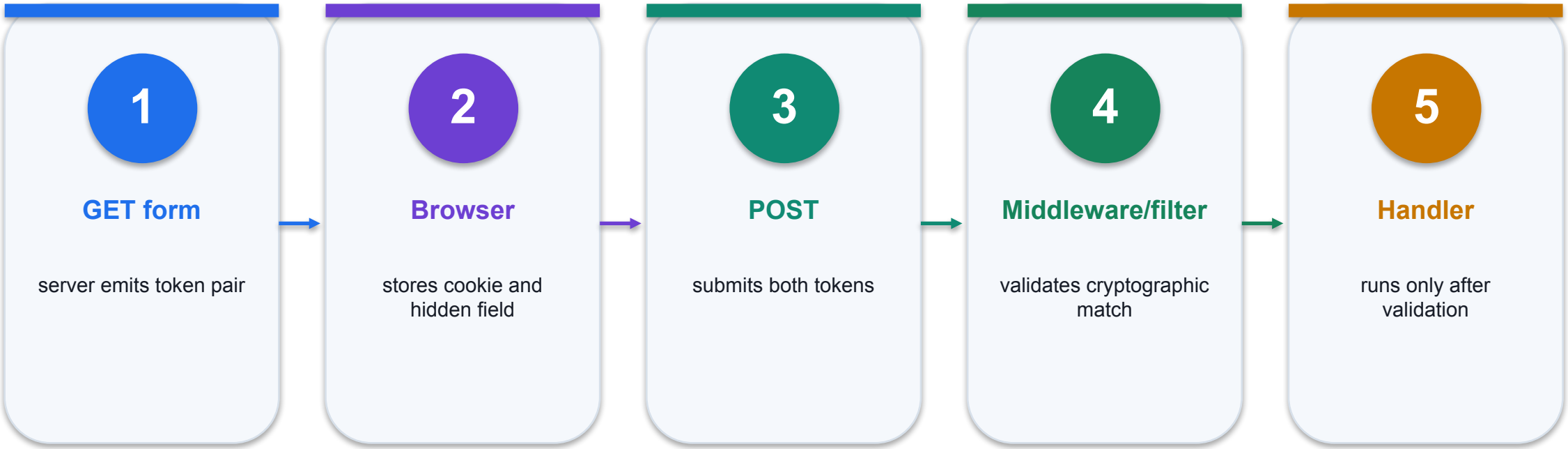
## ACCEPTANCE EVIDENCE

- ✓ Rendered label for equals input id
- ✓ Invalid POST redisplays message
- ✓ Generated form contains antiforgery token

**Record URL · command · status · build SHA**


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# Antiforgery request validation




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Antiforgery request validation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <form method="post"> <!-- FormTagHelper emits token -->
2   <button type="submit">Create</button>
3 </form>
```

1 **Cookie**
bound to browser session context

2 **Request token**
hidden form value/header

3 **SameSite**
reduces cross-site cookie sending

4 **POST/PUT/DELETE**
unsafe methods require protection

K3 / A3

Full procedures: Learner Guide


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# Antiforgery request validation · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| Cookie | bound to browser session context | Where is this enforced? |
| Request token | hidden form value/header | What is observable? |
| SameSite | reduces cross-site cookie sending | Who owns the value? |
| POST/PUT/DELETE | unsafe methods require protection | What rejects invalid input? |
| API auth | bearer-token APIs use different CSRF analysis | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Antiforgery request validation · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **400** | token pair is absent or invalid |
| **Token disabled** | cross-site form can execute command |
| **Mixed auth** | cookie API assumed bearer semantics |
| **Cached form** | stale token/cookie pair no longer matches |

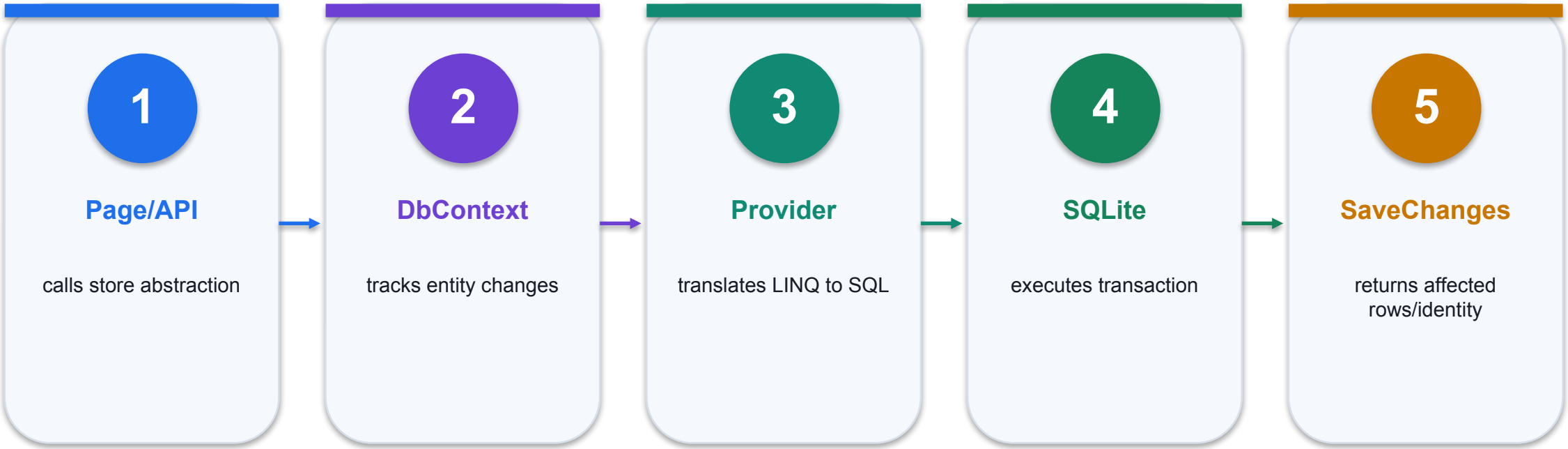
## ACCEPTANCE EVIDENCE

- ✓ POST without token returns 400
- ✓ Normal browser POST succeeds
- ✓ Unsafe cookie-auth endpoints are inventoried

**Record URL · command · status · build SHA**


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# EF Core SQLite persistence




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# EF Core SQLite persistence · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddDbContext<CrmDbContext>(o =>
2
   o.UseSqlite(builder.Configuration.GetConnectionString("LocalC
   RM")));
```

1 **DbSet**
query and mutation entry point

2 **Migration**
versioned schema change

3 **AsNoTracking**
read-only query optimisation

4 **Async**
does not block while I/O waits

K3 / A3

Full procedures: Learner Guide


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# EF Core SQLite persistence · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| DbSet | query and mutation entry point | Where is this enforced? |
| Migration | versioned schema change | What is observable? |
| AsNoTracking | read-only query optimisation | Who owns the value? |
| Async | does not block while I/O waits | What rejects invalid input? |
| Constraint | database protects required/unique rules | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# EF Core SQLite persistence · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **No database** | migration was never applied |
| **Locked** | long SQLite transaction blocks writer |
| **N+1** | navigation query issues repeated SQL |
| **Schema drift** | manual DB change bypassed migration |

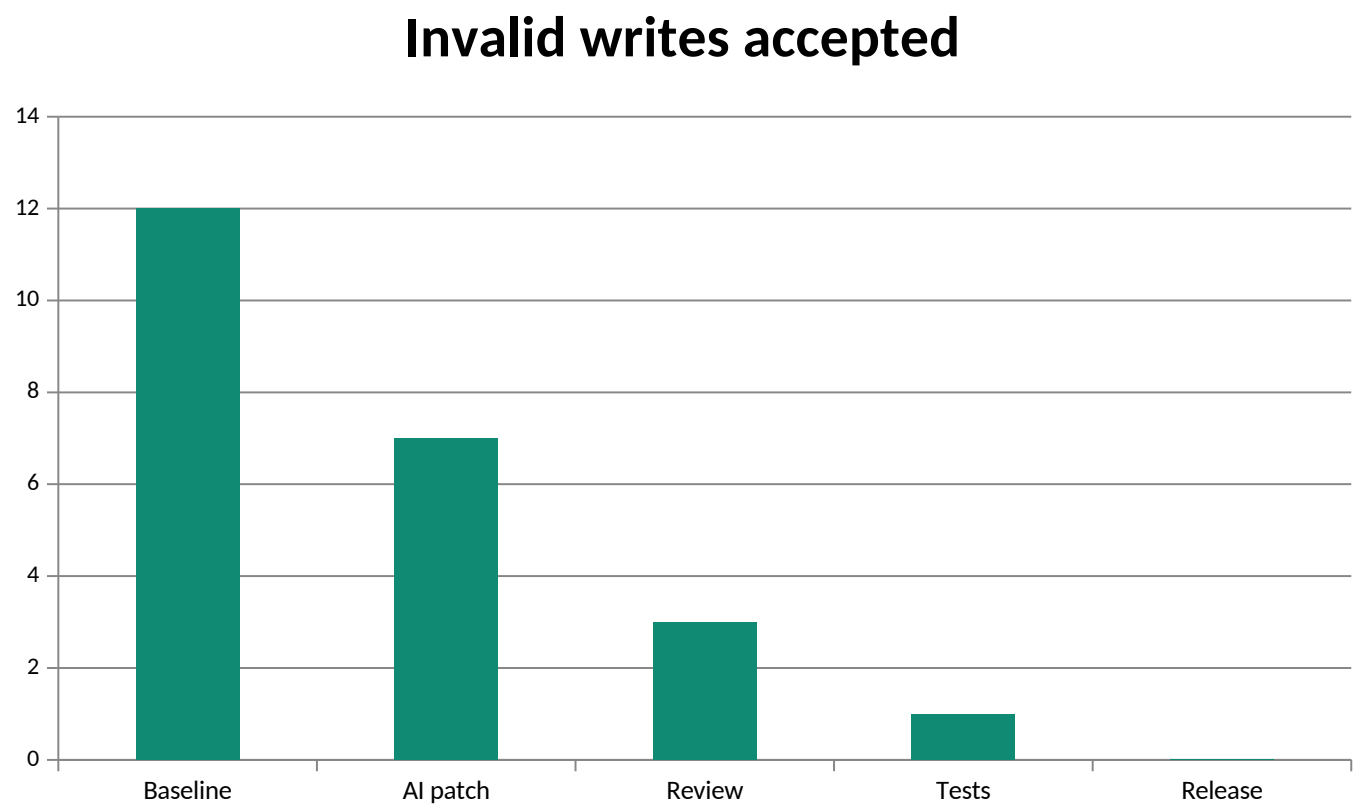
## ACCEPTANCE EVIDENCE

- ✓ dotnet ef database update succeeds
- ✓ Create survives application restart
- ✓ Integration test uses an isolated database

**Record URL · command · status · build SHA**


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# LO3 evidence profile




WHAT THE DATA SHOWS

**Server validation and antiforgery close unsafe input paths.**

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

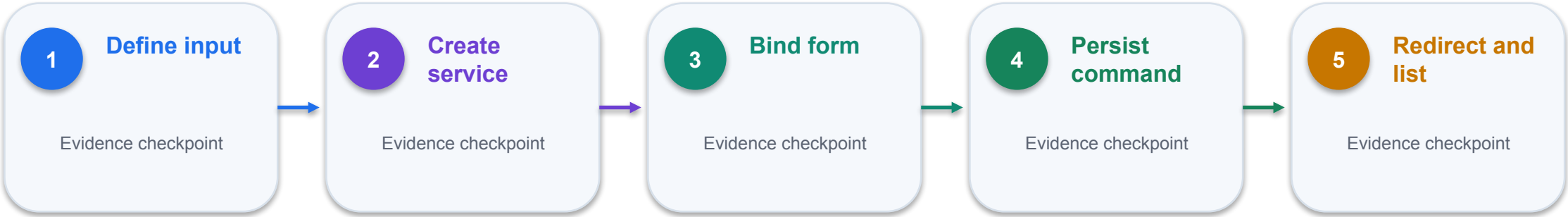
EDITABLE CHART


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65 MIN

# Activity 5: Create and List CRM Contacts with Razor Pages

Implement page-focused create/list handlers through an injectable contact store.




**YOU'LL PRODUCE** **input model, IContactStore boundary, create/list PageModels and successful PRG navigation**

Local snapshot: activities/activity-05-razor-create-list/src/LocalCRM · full commands in the Learner Guide


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# Activity 5 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Implement REQ-001 and REQ-002 with Razor Pages. Introduce IContactStore and an in-memory implementation registered as singleton for this temporary Activity only. Use a dedicated NewContact input model and Post/Redirect/Get. Limit changes to Models, Services, Pages/Contacts and Program.cs.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 5 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: input model, IContactStore boundary, create/list PageModels and successful PRG navigation.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

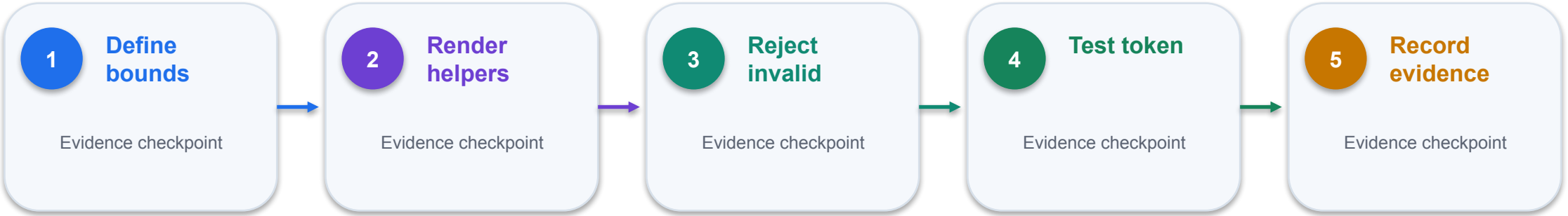
**K3/A3 EVIDENCE**


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50 MIN

# Activity 6: Enforce Validation and Antiforgery Controls

Reject invalid or forged commands while returning accessible recovery guidance.




**YOU'LL PRODUCE** **bounded DataAnnotations, field/summary errors, antiforgery proof and negative-path tests**

Local snapshot: activities/activity-06-validation-and-antiforgery/src/LocalCRM · full commands in the Learner Guide


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# Activity 6 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Harden the Create Razor Page. Name is required (2–80 characters), Email is required and valid, and Stage is Lead, Qualified or Customer. Add accessible field errors and validation summary. Preserve automatic antiforgery and add tests for invalid input and missing token.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 6 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: bounded DataAnnotations, field/summary errors, antiforgery proof and negative-path tests.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K3/A3 EVIDENCE**


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LO3 · K3 / A3

# Topic 3 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.


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TOPIC 4 · LO4 · K4 / A4

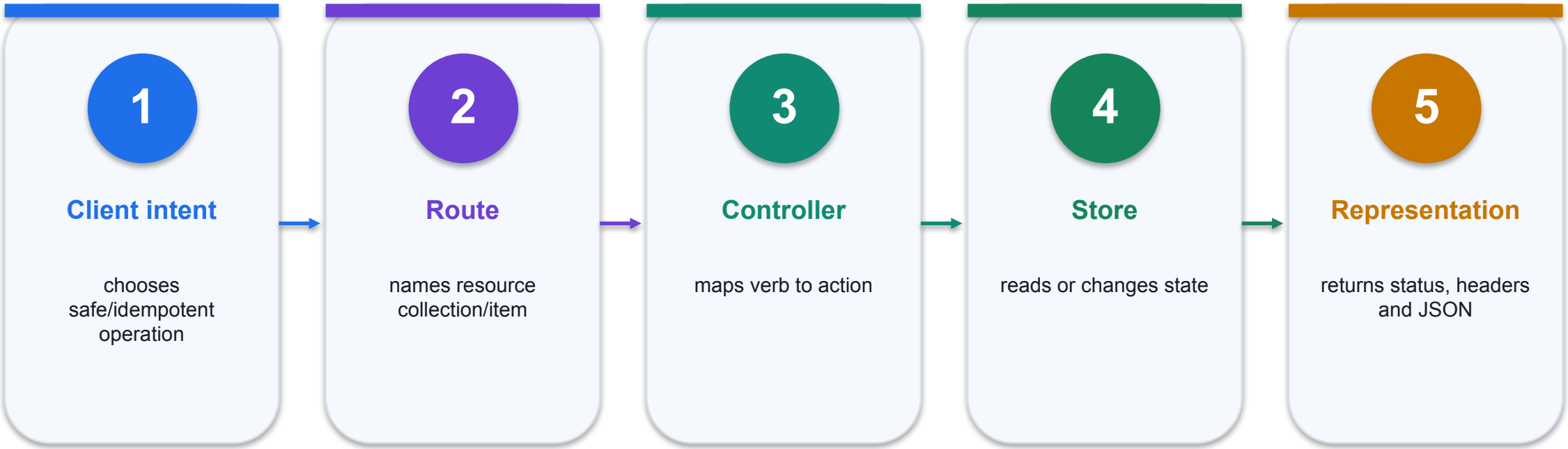
04

# Controller APIs and Component Interoperability

resource contract → controller boundary → persistence → observable HTTP evidence


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# Resource-oriented HTTP semantics




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Resource-oriented HTTP semantics · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [ApiController]
2 [Route("api/contacts")]
3 public sealed class ContactsController : ControllerBase { }
```

1. **GET /api/contacts**
   list; safe and idempotent
2. **POST /api/contacts**
   create; not inherently idempotent
3. **GET /api/contacts/{id}**
   read one
4. **PUT /api/contacts/{id}**
   replace; idempotent

K4 / A4

Full procedures: Learner Guide


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# Resource-oriented HTTP semantics · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| GET /api/contacts | list; safe and idempotent | Where is this enforced? |
| POST /api/contacts | create; not inherently idempotent | What is observable? |
| GET /api/contacts/{id} | read one | Who owns the value? |
| PUT /api/contacts/{id} | replace; idempotent | What rejects invalid input? |
| DELETE /api/contacts/{id} | remove; idempotent outcome | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Resource-oriented HTTP semantics · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Verb tunnel** | all commands use POST without semantics |
| **Noun mismatch** | route describes implementation, not resource |
| **GET mutates** | caches/retries can change state |
| **No Location** | client cannot find created resource |

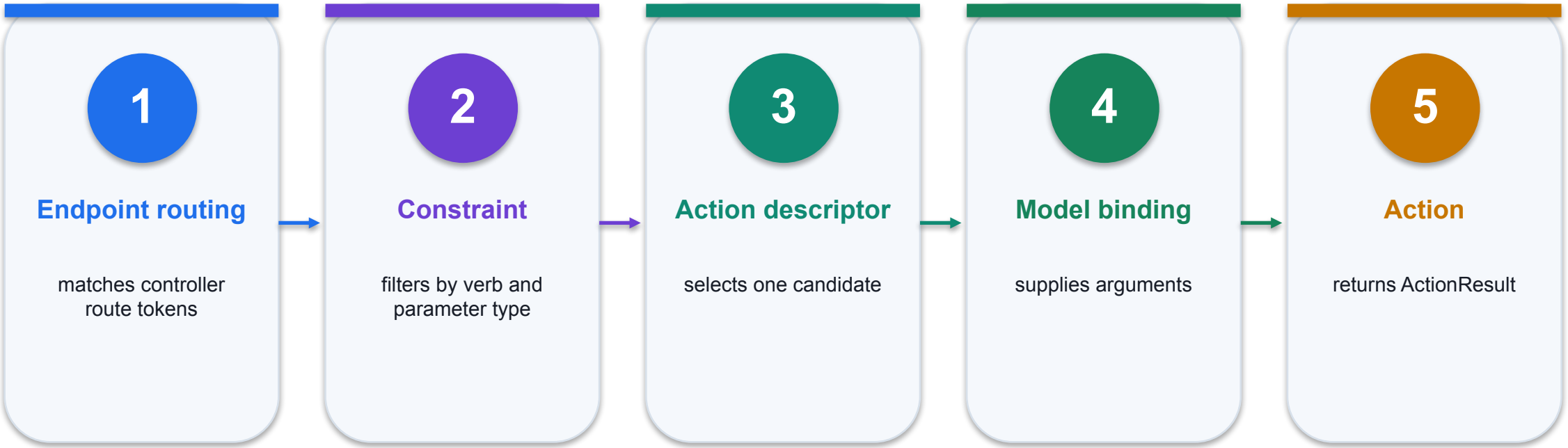
## ACCEPTANCE EVIDENCE

- ✓ OpenAPI lists five intended operations
- ✓ GET never changes row count
- ✓ POST returns 201 and Location header

**Record URL · command · status · build SHA**


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# Controller routing and action selection




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Controller routing and action selection · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [HttpGet("{id:int}", Name = "GetContact")]
2 public async Task<ActionResult<ContactDto>> Get(int id) {
  /*...*/ }
```

**1 [Route]**
controller-level path prefix

**2 [HttpGet("{id:int}")]**
verb plus constrained template

**3 [FromRoute]**
explicit path source

**4 CreatedAtAction**
generates canonical item URL

K4 / A4

Full procedures: Learner Guide


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# Controller routing and action selection · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| [Route] | controller-level path prefix | Where is this enforced? |
| [HttpGet("{id:int}")] | verb plus constrained template | What is observable? |
| [FromRoute] | explicit path source | Who owns the value? |
| CreatedAtAction | generates canonical item URL | What rejects invalid input? |
| Name | supports stable link generation | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Controller routing and action selection · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **AmbiguousMatch** | two actions share equivalent route |
| **404** | constraint rejects supplied id |
| **Wrong URL** | CreatedAtAction route values do not match |
| **Leaked action** | public method lacks NonAction/verb attribute |

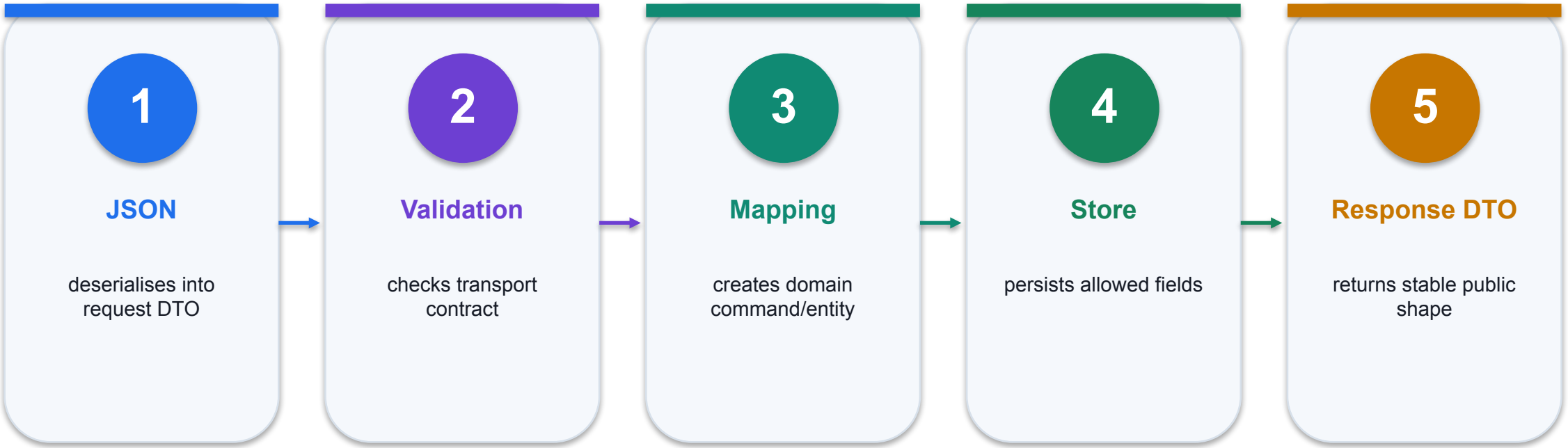
## ACCEPTANCE EVIDENCE

- ✓ Route table has no duplicate method+path
- ✓ id=abc produces 404
- ✓ Location URL resolves to the new representation

**Record URL · command · status · build SHA**


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# DTO and domain boundary




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# DTO and domain boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 public sealed record CreateContactDto(
2   [Required, StringLength(80)] string Name,
3   [Required, EmailAddress] string Email, string Stage);
```

1 **CreateContactDto**
client-settable fields only

2 **ContactDto**
id/status/timestamps safe to expose

3 **Entity**
persistence and domain state

4 **Mapping**
explicit code makes changes reviewable

K4 / A4

Full procedures: Learner Guide


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# DTO and domain boundary · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| CreateContactDto | client-settable fields only | Where is this enforced? |
| ContactDto | id/status/timestamps safe to expose | What is observable? |
| Entity | persistence and domain state | Who owns the value? |
| Mapping | explicit code makes changes reviewable | What rejects invalid input? |
| Versioning | contract change is deliberate | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# DTO and domain boundary · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Mass assignment** | entity is accepted directly |
| **Circular JSON** | navigation properties leak |
| **Contract drift** | database rename breaks clients |
| **Sensitive field** | internal note appears in response |

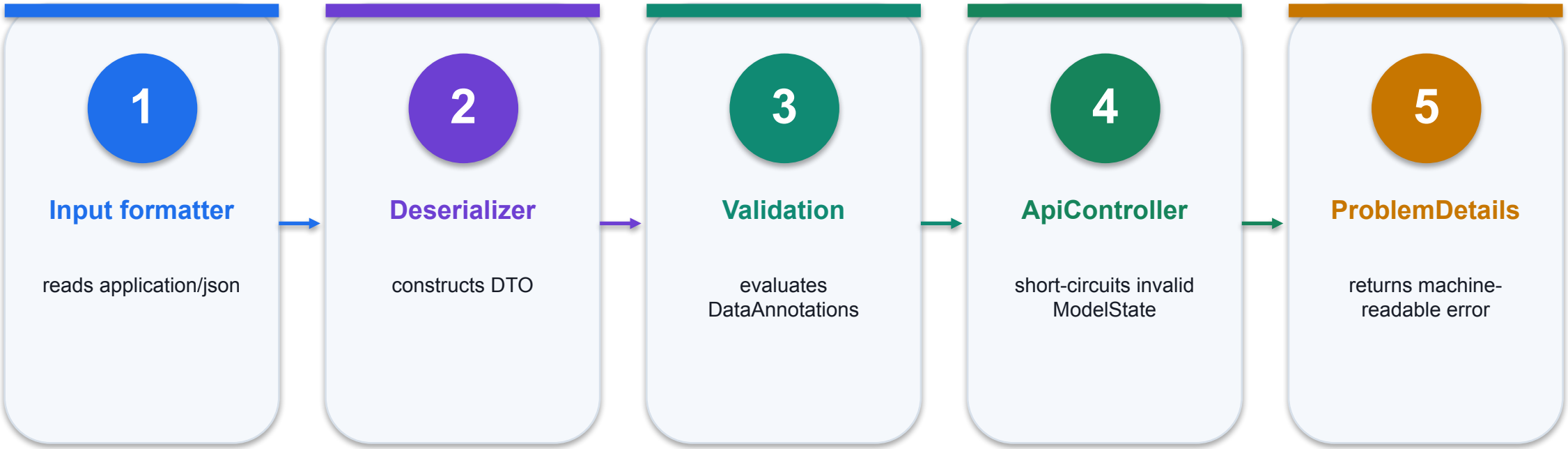
## ACCEPTANCE EVIDENCE

- ✓ Unknown JSON property is handled by policy
- ✓ Response excludes internal fields
- ✓ Mapping test covers every public property

**Record URL · command · status · build SHA**


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# API model binding and automatic 400




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# API model binding and automatic 400 · worked artifact

REVIEWED EXCERPT · .NET 10

```
curl -i -X POST /api/contacts \
+ -H "Content-Type: application/json" -d "{\"summary\":\"\"}"
```

1. **Content-Type**
   must select JSON input formatter
2. **400**
   syntactically valid but contract-invalid
3. **415**
   unsupported media type
4. **ValidationProblemDetails**
   errors keyed by field

K4 / A4

Full procedures: Learner Guide


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# API model binding and automatic 400 · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| Content-Type | must select JSON input formatter | Where is this enforced? |
| 400 | syntactically valid but contract-invalid | What is observable? |
| 415 | unsupported media type | Who owns the value? |
| ValidationProblemDetails | errors keyed by field | What rejects invalid input? |
| Nullable | requiredness should be explicit | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 4 · DIAGNOSE + VERIFY

# API model binding and automatic 400 · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **400 hidden** | client ignores ProblemDetails body |
| **415** | Content-Type header is missing/wrong |
| **Empty object** | nullable/default contract too permissive |
| **Exception** | custom binder throws instead of reporting error |

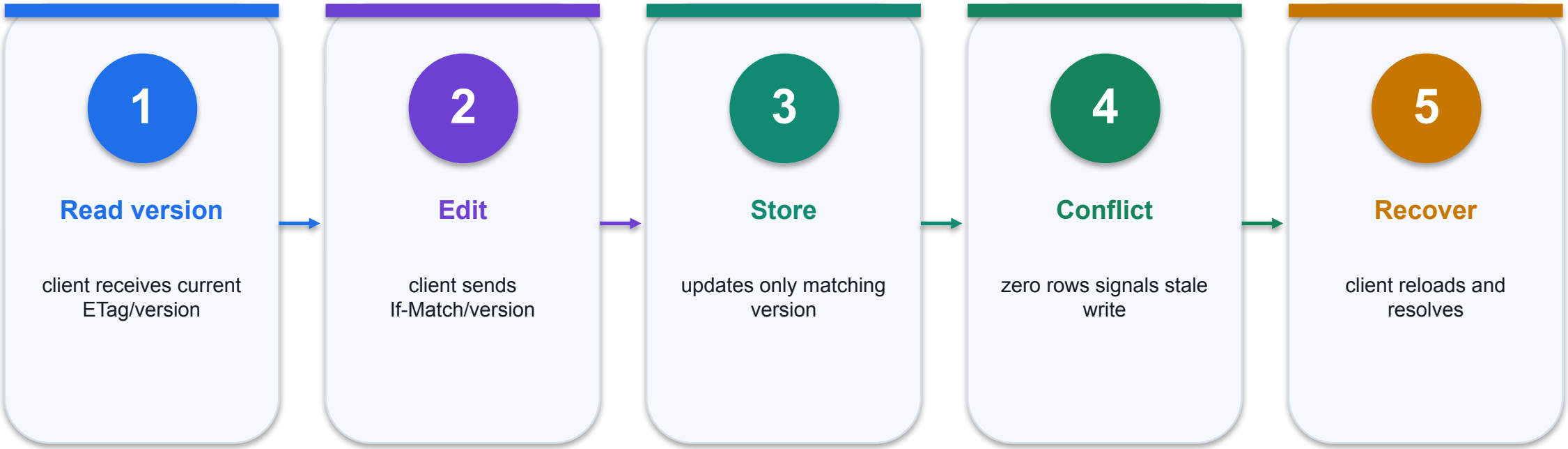
## ACCEPTANCE EVIDENCE

- ✓ Invalid DTO returns application/problem+json
- ✓ Error dictionary names summary
- ✓ Controller action is not invoked

**Record URL · command · status · build SHA**


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# CRUD status and concurrency contract




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# CRUD status and concurrency contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 if (dto.Version != entity.Version)
2   return Conflict(new ProblemDetails { Title = "Stale
  contact" });
```

1. **200** — successful read/update with body
2. **201** — created with Location
3. **204** — successful delete without body
4. **404** — resource absent

K4 / A4

Full procedures: Learner Guide


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TOPIC 4 · DESIGN DETAIL

# CRUD status and concurrency contract · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| 200 | successful read/update with body | Where is this enforced? |
| 201 | created with Location | What is observable? |
| 204 | successful delete without body | Who owns the value? |
| 404 | resource absent | What rejects invalid input? |
| 409/412 | state conflict/precondition failed | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# CRUD status and concurrency contract · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Lost update** | last writer silently overwrites |
| **Wrong 200** | missing resource returns success |
| **Duplicate create** | retry creates second record |
| **204 with body** | response violates its status semantics |

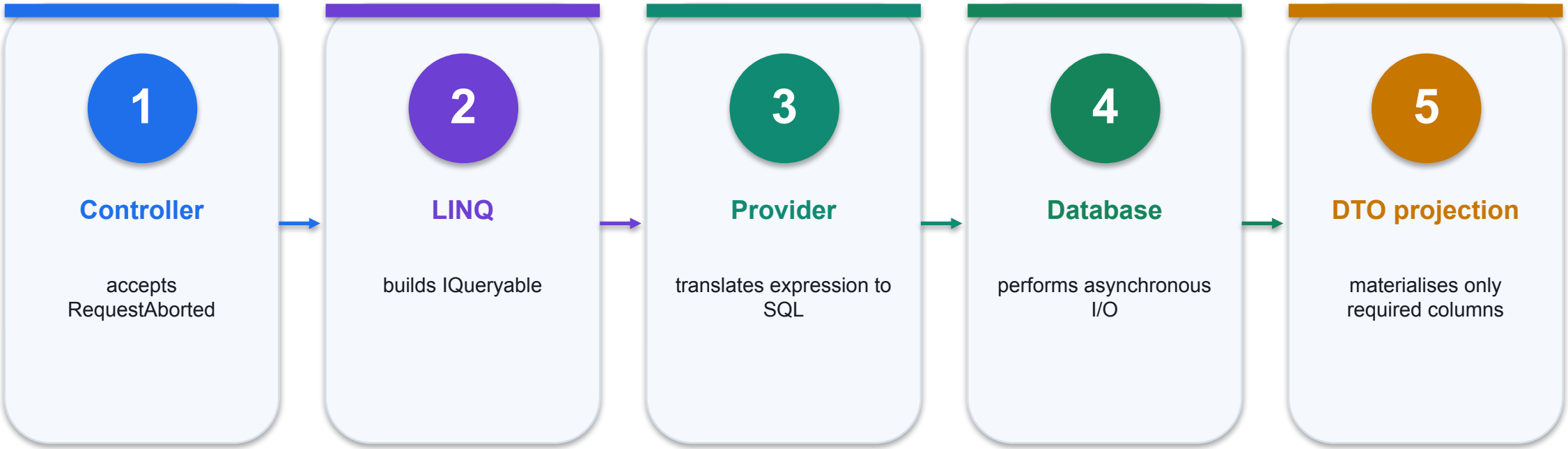
## ACCEPTANCE EVIDENCE

- ✓ Stale update produces 409/412
- ✓ Delete missing id produces 404
- ✓ Create provides resolvable Location

**Record URL · command · status · build SHA**


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# Async EF Core API path




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Async EF Core API path · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 var rows = await db.Contacts.AsNoTracking()
2  .Select(x => new ContactDto(x.Id, x.Name, x.Email, x.Stage))
3  .ToListAsync(ct);
```

1 **ToListAsync**
executes query asynchronously

2 **Select**
projects before materialisation

3 **AsNoTracking**
avoids change tracker for reads

4 **CancellationToken**
stops abandoned work

K4 / A4

Full procedures: Learner Guide


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# Async EF Core API path · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| ToListAsync | executes query asynchronously | Where is this enforced? |
| Select | projects before materialisation | What is observable? |
| AsNoTracking | avoids change tracker for reads | Who owns the value? |
| CancellationToken | stops abandoned work | What rejects invalid input? |
| SaveChangesAsync | persists transaction | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Async EF Core API path · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Client eval** | expression cannot translate |
| **Over-fetch** | entity graph is loaded unnecessarily |
| **No cancellation** | work continues after disconnect |
| **Parallel DbContext** | same context is used concurrently |

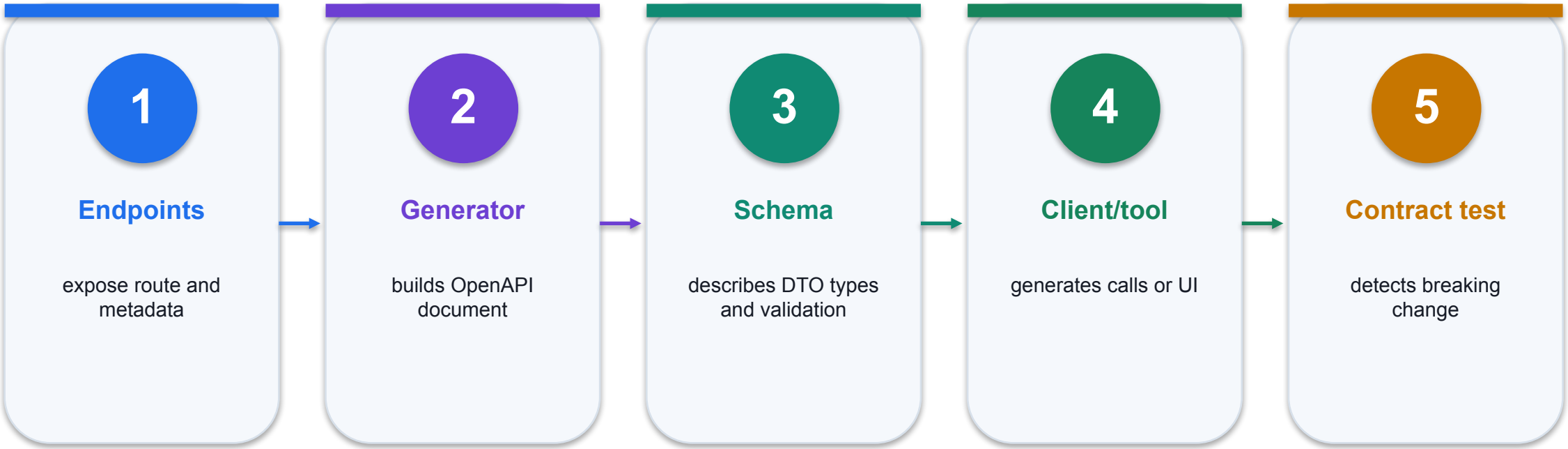
## ACCEPTANCE EVIDENCE

- ✓ SQL log selects only DTO columns
- ✓ Cancelled request ends query
- ✓ Load test shows no sync blocking hotspot

**Record URL · command · status · build SHA**


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# OpenAPI as interoperability contract




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# OpenAPI as interoperability contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 builder.Services.AddOpenApi();
2 if (app.Environment.IsDevelopment()) app.MapOpenApi();
```

1 **paths**
method and route operations

2 **operationId**
stable generated-client method

3 **components/schemas**
reusable JSON shapes

4 **responses**
document every intended status

K4 / A4

Full procedures: Learner Guide


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# OpenAPI as interoperability contract · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| paths | method and route operations | Where is this enforced? |
| operationId | stable generated-client method | What is observable? |
| components/schemas | reusable JSON shapes | Who owns the value? |
| responses | document every intended status | What rejects invalid input? |
| securitySchemes | declares authentication mechanism | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# OpenAPI as interoperability contract · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Undocumented status** | client cannot model failure |
| **Schema leak** | entity exposes internal fields |
| **Breaking rename** | generated client no longer compiles |
| **Prod exposure** | interactive docs reveal attack surface |

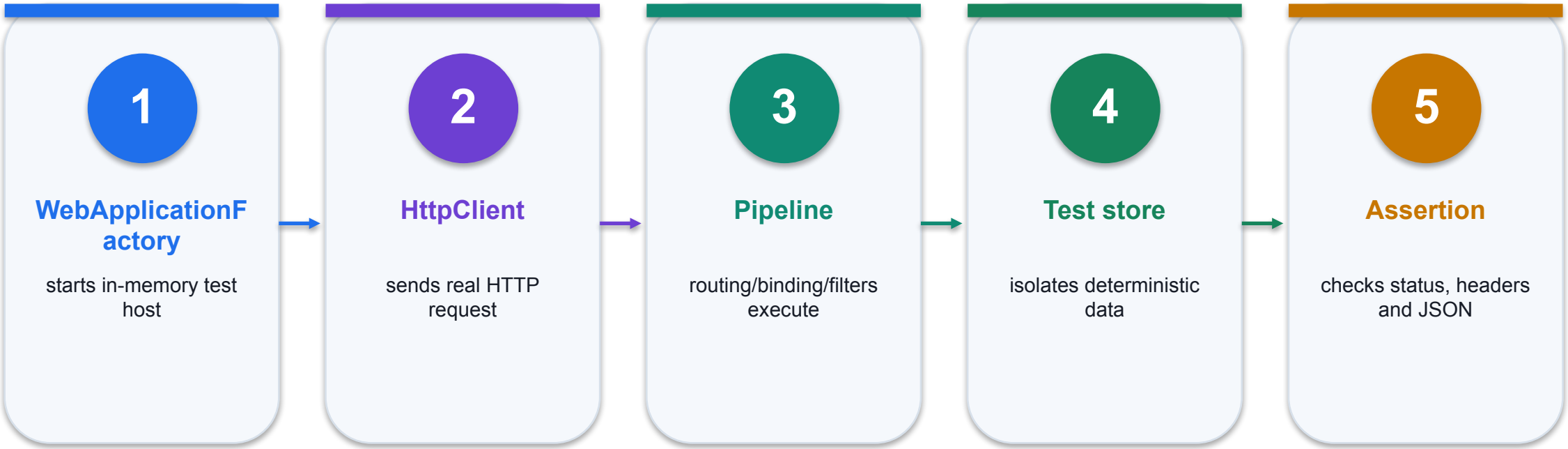
## ACCEPTANCE EVIDENCE

- ✓ openapi.json validates
- ✓ Snapshot diff contains no accidental breaking change
- ✓ 201/400/404 schemas are declared

**Record URL · command · status · build SHA**


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# Integration tests and ProblemDetails




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Integration tests and ProblemDetails · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  var response = await client.PostAsJsonAsync("/api/contacts",
   new { summary = "" });
2  Assert.Equal(HttpStatusCode.BadRequest, response.StatusCode);
```

1 **Arrange**
seed isolated database

2 **Act**
call public endpoint only

3 **Assert**
status + content type + shape

4 **ProblemDetails**
stable error envelope

K4 / A4

Full procedures: Learner Guide


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# Integration tests and ProblemDetails · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Arrange** | seed isolated database | Where is this enforced? |
| **Act** | call public endpoint only | What is observable? |
| **Assert** | status + content type + shape | Who owns the value? |
| **ProblemDetails** | stable error envelope | What rejects invalid input? |
| **Trace id** | links client failure to server log | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Integration tests and ProblemDetails · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Unit-only** | middleware/routing regression is invisible |
| **Shared DB** | tests affect each other |
| **Body-only** | wrong status still passes |
| **Stack trace leak** | error response exposes internals |

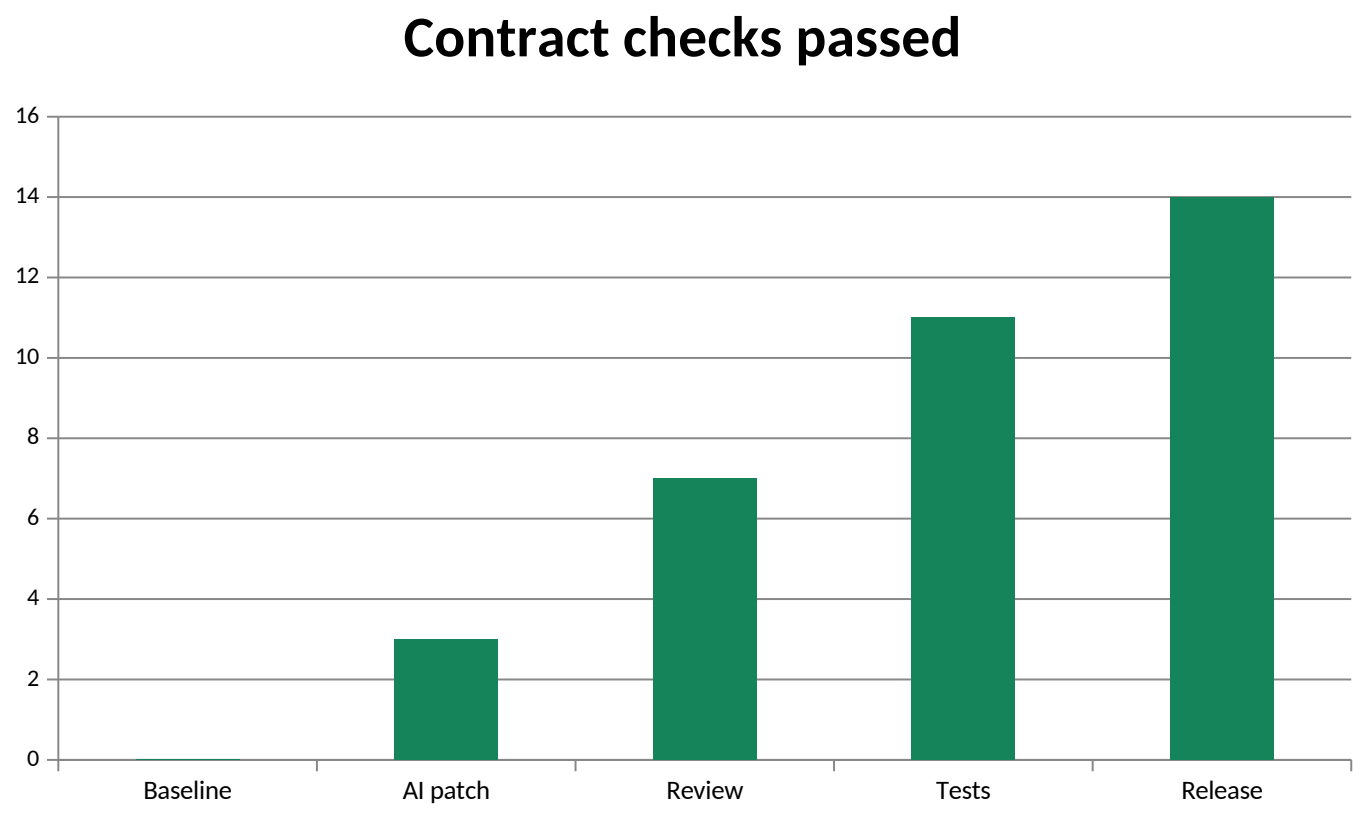
## ACCEPTANCE EVIDENCE

- ✓ Happy and invalid paths run through HTTP
- ✓ ProblemDetails contains traceId
- ✓ Tests pass in parallel with isolated state

**Record URL · command · status · build SHA**


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# LO4 evidence profile




WHAT THE DATA SHOWS

**Interoperability confidence comes from status, header and schema assertions.**

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

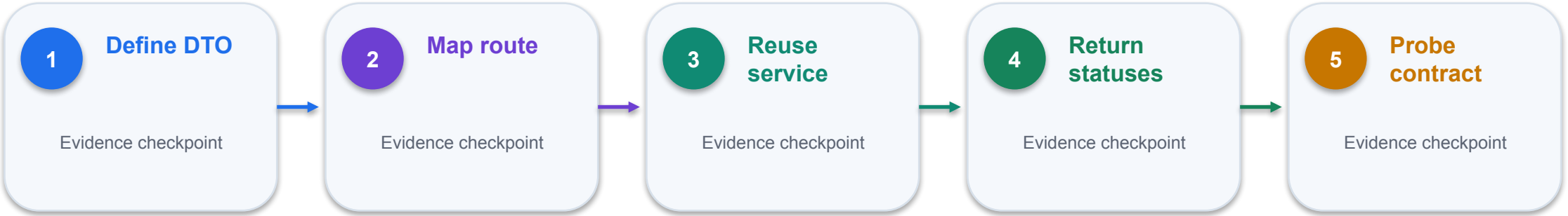
EDITABLE CHART


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65 MIN

# Activity 7: Expose a Controller-Based CRUD API

Add a JSON controller surface that reuses the application service without exposing persistence entities.




**YOU'LL PRODUCE** **controller routes, DTOs, 201/200/400/404 behaviour and curl evidence**

Local snapshot: activities/activity-07-controller-api/src/LocalCRM · full commands in the Learner Guide


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# Activity 7 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Add an ApiController at /api/contacts that supports GET collection, GET item and POST create. Use transport DTOs, IContactStore, CreatedAtAction and CancellationToken. Do not return persistence entities directly. Include 400 and 404 evidence.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 7 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: controller routes, DTOs, 201/200/400/404 behaviour and curl evidence.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K4/A4 EVIDENCE**


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70 MIN

# Activity 8: Prove Persistence and Interoperability

Replace volatile state with EF Core SQLite, publish an OpenAPI contract and verify components through integration tests.




**YOU'LL PRODUCE** **SQLite migration, persistent CRUD path, openapi.json and HTTP integration tests**

Local snapshot: activities/activity-08-sqlite-openapi-tests/src/LocalCRM · full commands in the Learner Guide


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# Activity 8 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Replace InMemoryContactStore with EF Core SQLite behind IContactStore. Add a migration, async cancellation-aware queries, OpenAPI in Development, and WebApplicationFactory tests for 201, 400 and 404. Preserve existing page and API contracts.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 8 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: SQLite migration, persistent CRUD path, openapi.json and HTTP integration tests.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K4/A4 EVIDENCE**


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LO4 · K4 / A4

# Topic 4 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.


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TOPIC 5 · LO5 · K5 / A5

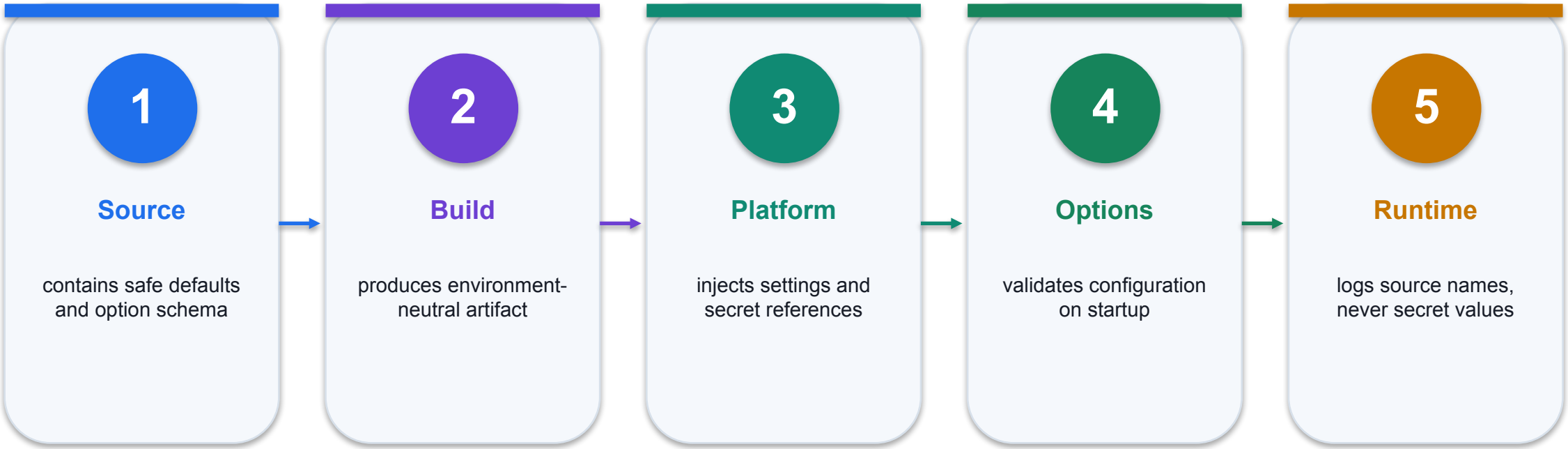
05

# Production Hosting, Azure and Design Evidence

deployable artifact → observable service → safe release → traceable design record


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# Environment-specific configuration and secrets




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Environment-specific configuration and secrets · worked artifact

REVIEWED EXCERPT · .NET 10

```
var cs =
builder.Configuration.GetConnectionString("LocalCRM")
  ?? throw new InvalidOperationException("Missing LocalCRM
connection");
```

1. **ASPNETCORE_ENVIRONMENT**
   Development/Staging/Production
2. **App__FeatureFlag**
   environment variable key encoding
3. **Key Vault reference**
   secret stays outside app settings
4. **ValidateOnStart**
   fails before traffic

K5 / A5

Full procedures: Learner Guide


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# Environment-specific configuration and secrets · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| ASPNETCORE_ENVIRONMENT | Development/Staging/Production | Where is this enforced? |
| App__FeatureFlag | environment variable key encoding | What is observable? |
| Key Vault reference | secret stays outside app settings | Who owns the value? |
| ValidateOnStart | fails before traffic | What rejects invalid input? |
| Rotation | changes secret without source edit | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Environment-specific configuration and secrets · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Secret in repo** | credential entered appsettings.json |
| **Wrong slot value** | setting was not marked deployment-slot specific |
| **Silent default** | production uses developer fallback |
| **Log leak** | configuration object was serialised |

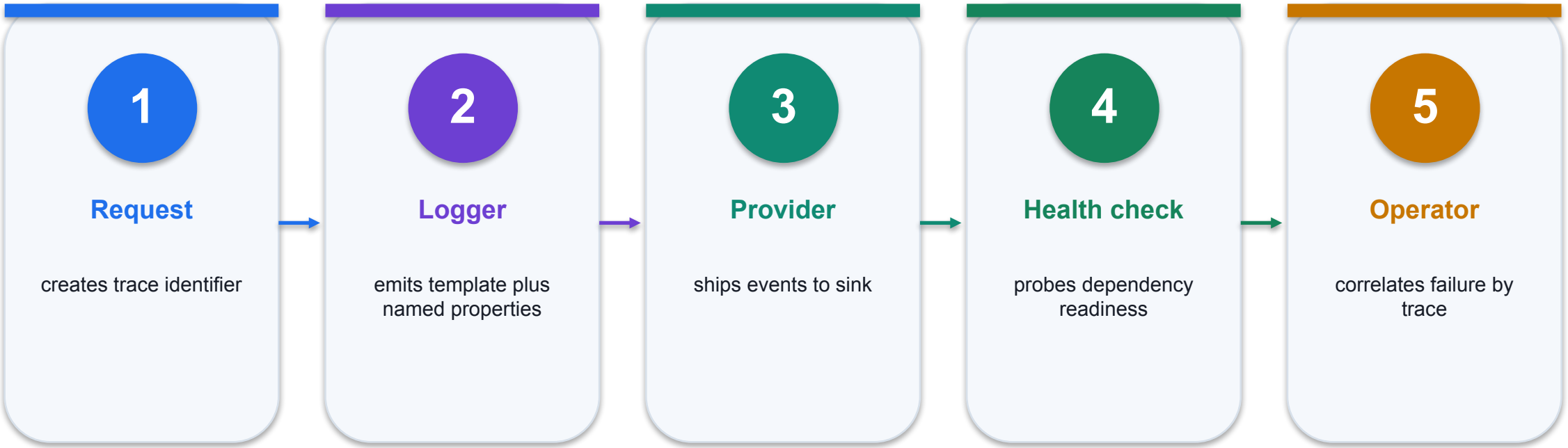
## ACCEPTANCE EVIDENCE

- ✓ Secret scan is clean
- ✓ Production start fails without required setting
- ✓ Rotated value is picked up via platform process

**Record URL · command · status · build SHA**


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# Structured logging and health checks




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Structured logging and health checks · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 builder.Services.AddHealthChecks().AddDbContextCheck<CrmDbCon
  text>();
2 app.MapHealthChecks("/health/ready");
```

1 **Template**
Contact {ContactId} created

2 **Level**
Information/Warning/Error chosen by action

3 **Liveness**
process is running

4 **Readiness**
dependencies can serve traffic

K5 / A5

Full procedures: Learner Guide


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# Structured logging and health checks · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| Template | Contact {ContactId} created | Where is this enforced? |
| Level | Information/Warning/Error chosen by action | What is observable? |
| Liveness | process is running | Who owns the value? |
| Readiness | dependencies can serve traffic | What rejects invalid input? |
| PII | not emitted as log property | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Structured logging and health checks · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **String log** | fields cannot be queried |
| **False green** | liveness ignores failed database |
| **PII leak** | request body is logged |
| **No trace** | client error cannot be correlated |

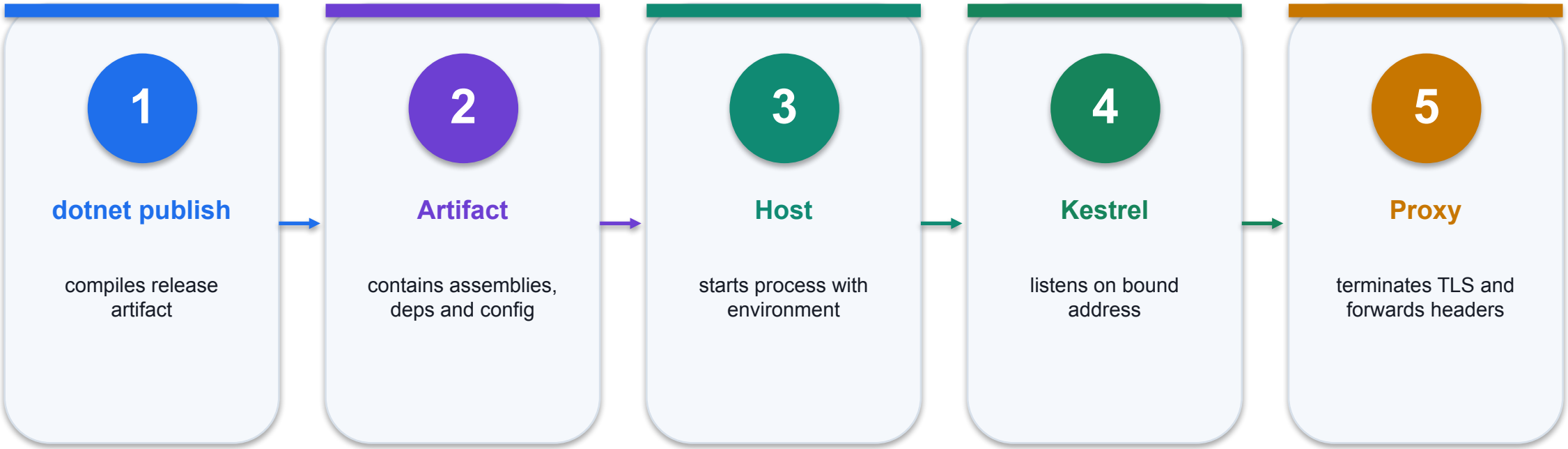
## ACCEPTANCE EVIDENCE

- ✓ /health/live and /health/ready have distinct meaning
- ✓ 500 log carries trace id
- ✓ No sensitive payload appears in sink

**Record URL · command · status · build SHA**


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# Publish output and hosting boundary




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Publish output and hosting boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet publish -c Release -o ./publish
2 ASPNETCORE_URLS=http://+:8080 dotnet ./publish/LocalCRM.dll
```

1. **Framework-dependent**
   small artifact requires installed runtime
2. **Self-contained**
   larger artifact bundles runtime
3. **ForwardedHeaders**
   trust only known proxy
4. **PORT/URLS**
   platform supplies binding

K5 / A5

Full procedures: Learner Guide


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# Publish output and hosting boundary · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| **Framework-dependent** | small artifact requires installed runtime | Where is this enforced? |
| **Self-contained** | larger artifact bundles runtime | What is observable? |
| **ForwardedHeaders** | trust only known proxy | Who owns the value? |
| **PORT/URLS** | platform supplies binding | What rejects invalid input? |
| **Shutdown** | SIGTERM allows graceful stop | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Publish output and hosting boundary · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **502** | process crashed or bound wrong port |
| **Redirect loop** | proxy headers are not handled |
| **Runtime missing** | framework-dependent target lacks .NET |
| **Partial artifact** | publish folder was copied inconsistently |

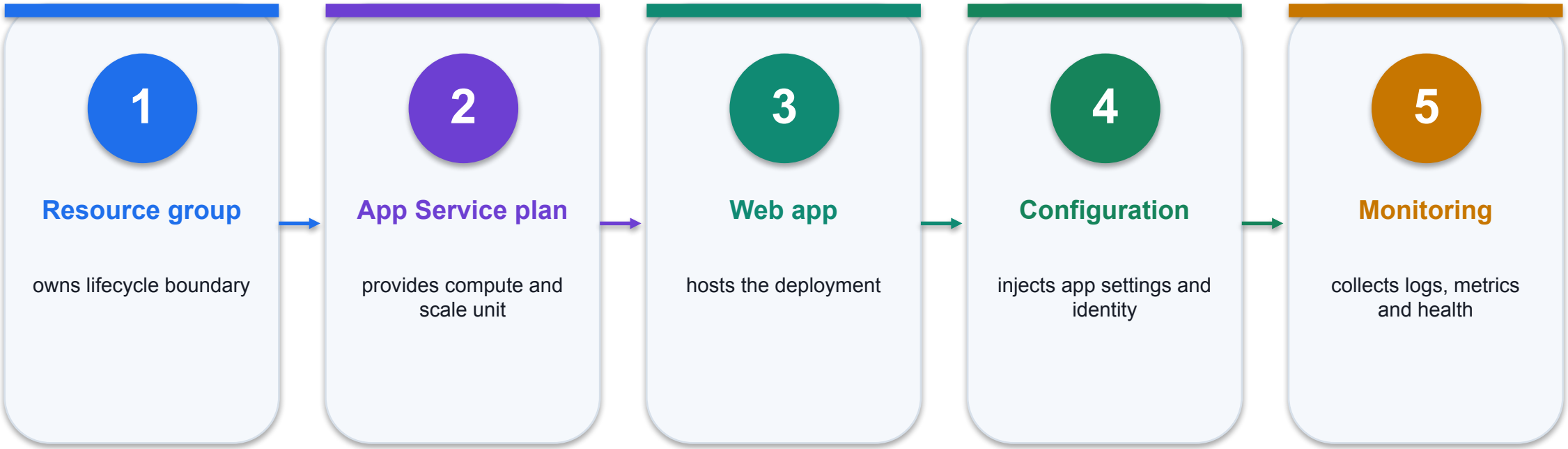
## ACCEPTANCE EVIDENCE

- ✓ Release artifact starts in a clean directory
- ✓ HTTP probe reaches bound port
- ✓ SIGTERM stops without dropping active request

**Record URL · command · status · build SHA**


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# Azure App Service component model




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Azure App Service component model · worked artifact

REVIEWED EXCERPT · .NET 10

```
az webapp config appsettings set --resource-group rg-localcrm \
+ --name localcrm-web --settings ASPNETCORE_ENVIRONMENT=Production
```

1. **Region** — latency/data residency decision
2. **Plan SKU** — cost, scale and feature envelope
3. **Linux/Windows** — runtime hosting choice
4. **Always On** — avoids idle unload where supported

K5 / A5

Full procedures: Learner Guide


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# Azure App Service component model · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| Region | latency/data residency decision | Where is this enforced? |
| Plan SKU | cost, scale and feature envelope | What is observable? |
| Linux/Windows | runtime hosting choice | Who owns the value? |
| Always On | avoids idle unload where supported | What rejects invalid input? |
| Health check | removes unhealthy instances | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Azure App Service component model · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Wrong plan** | required feature is unavailable |
| **Cold/idle** | Always On not enabled where needed |
| **Unhealthy instance** | health path not configured |
| **Configuration drift** | portal-only edits bypass design record |

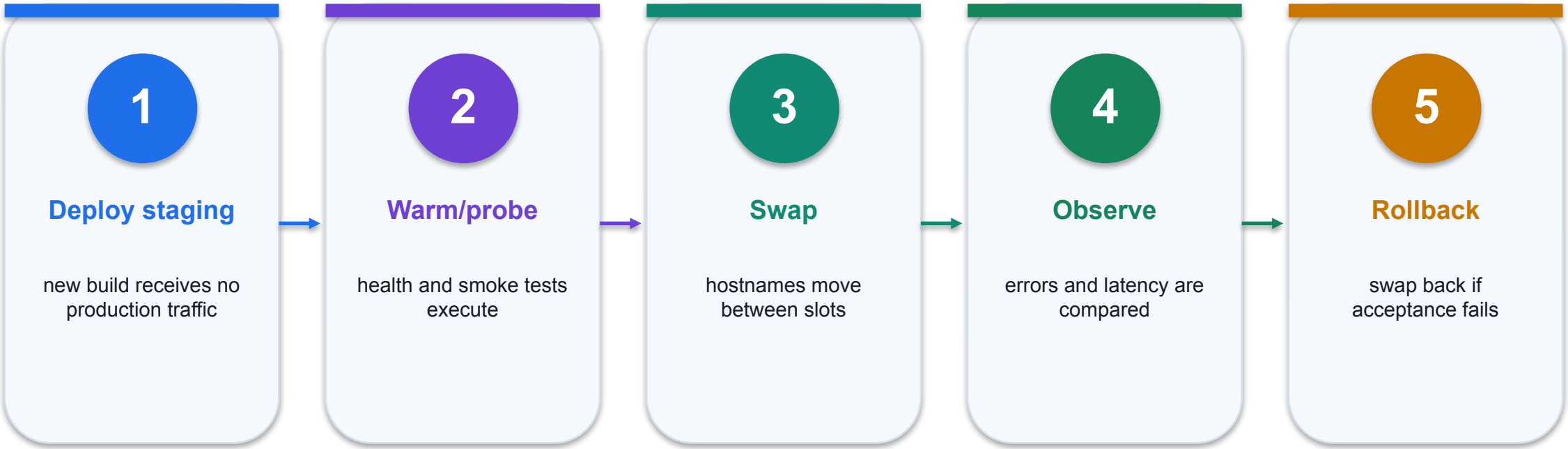
## ACCEPTANCE EVIDENCE

- ✓ Resource inventory names RG, plan and app
- ✓ Health check path is configured
- ✓ Deployment settings match documented environment

**Record URL · command · status · build SHA**


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# Deployment slots and safe promotion




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Deployment slots and safe promotion · worked artifact

REVIEWED EXCERPT · .NET 10

```
az webapp deployment slot swap -g rg-localcrm -n localcrm-web \
+ --slot staging --target-slot production
```

1. **Slot setting**
   secret/connection stays with slot
2. **Warm-up**
   application initialises before swap
3. **Preview**
   validates configuration changes
4. **Smoke test**
   uses public contract

K5 / A5

Full procedures: Learner Guide


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# Deployment slots and safe promotion · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| **Slot setting** | secret/connection stays with slot | Where is this enforced? |
| **Warm-up** | application initialises before swap | What is observable? |
| **Preview** | validates configuration changes | Who owns the value? |
| **Smoke test** | uses public contract | What rejects invalid input? |
| **Rollback** | predefined threshold and command | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# Deployment slots and safe promotion · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Bad swap** | slot-specific setting was not sticky |
| **Warm-up miss** | first production request compiles/initialises |
| **No rollback** | failed build remains live |
| **Schema incompatibility** | old app cannot run after migration |

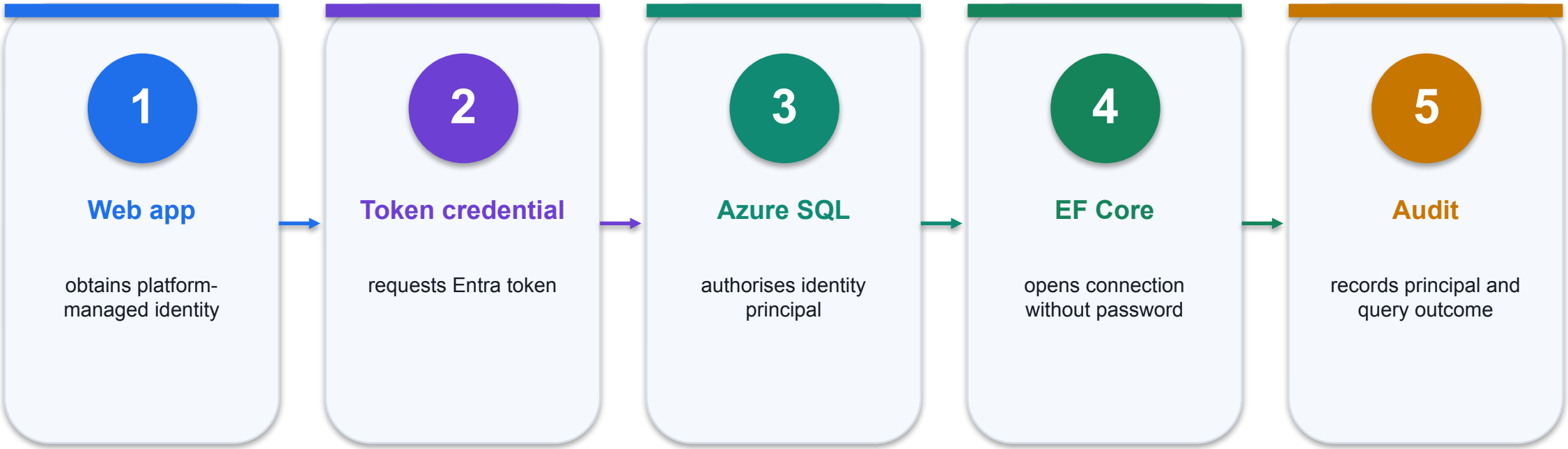
## ACCEPTANCE EVIDENCE

- ✓ Staging smoke test passes before swap
- ✓ Production health remains green
- ✓ Rollback step is rehearsed and timed

**Record URL · command · status · build SHA**


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# Managed identity and database access




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# Managed identity and database access · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddAzureClients(c => c.UseCredential(new
   DefaultAzureCredential()));
```

1 **System assigned**
identity lifecycle follows app

2 **User assigned**
identity can be shared/reused

3 **Least privilege**
grant only required database role

4 **DefaultAzureCredential**
local and hosted credential chain

K5 / A5

Full procedures: Learner Guide


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# Managed identity and database access · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| System assigned | identity lifecycle follows app | Where is this enforced? |
| User assigned | identity can be shared/reused | What is observable? |
| Least privilege | grant only required database role | Who owns the value? |
| DefaultAzureCredential | local and hosted credential chain | What rejects invalid input? |
| No password | rotation burden moves to platform | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 5 · DIAGNOSE + VERIFY

# Managed identity and database access · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Login failed** | identity lacks database user/role |
| **Wrong tenant** | credential resolved unexpected account |
| **Overprivilege** | app identity is database owner |
| **Local mismatch** | developer chain differs from hosted identity |

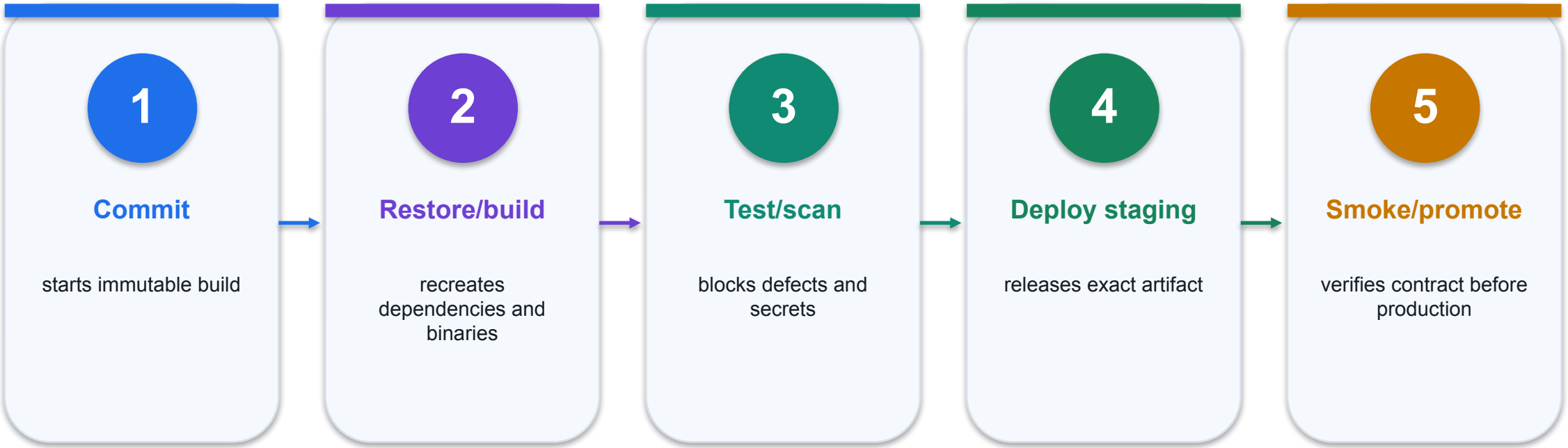
## ACCEPTANCE EVIDENCE

- ✓ App connects with passwordless identity
- ✓ Database audit names the app principal
- ✓ Denied operation proves least privilege

**Record URL · command · status · build SHA**


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# CI/CD gates and smoke evidence




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# CI/CD gates and smoke evidence · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet restore --locked-mode
2 dotnet build -c Release --no-restore
3 dotnet test -c Release --no-build
```

1 **Artifact SHA**
links deployed bytes to commit

2 **dotnet test**
unit and integration gate

3 **Dependency scan**
known-vulnerability control

4 **Environment approval**
separates build from promotion

K5 / A5

Full procedures: Learner Guide


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# CI/CD gates and smoke evidence · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
|---|---|---|
| **Artifact SHA** | links deployed bytes to commit | Where is this enforced? |
| **dotnet test** | unit and integration gate | What is observable? |
| **Dependency scan** | known-vulnerability control | Who owns the value? |
| **Environment approval** | separates build from promotion | What rejects invalid input? |
| **Smoke** | health plus one business transaction | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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# CI/CD gates and smoke evidence · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Mutable deploy** | server rebuilds from source |
| **No lock** | dependency version changes silently |
| **Green build/bad app** | no HTTP smoke test |
| **Untraceable** | artifact SHA not recorded |

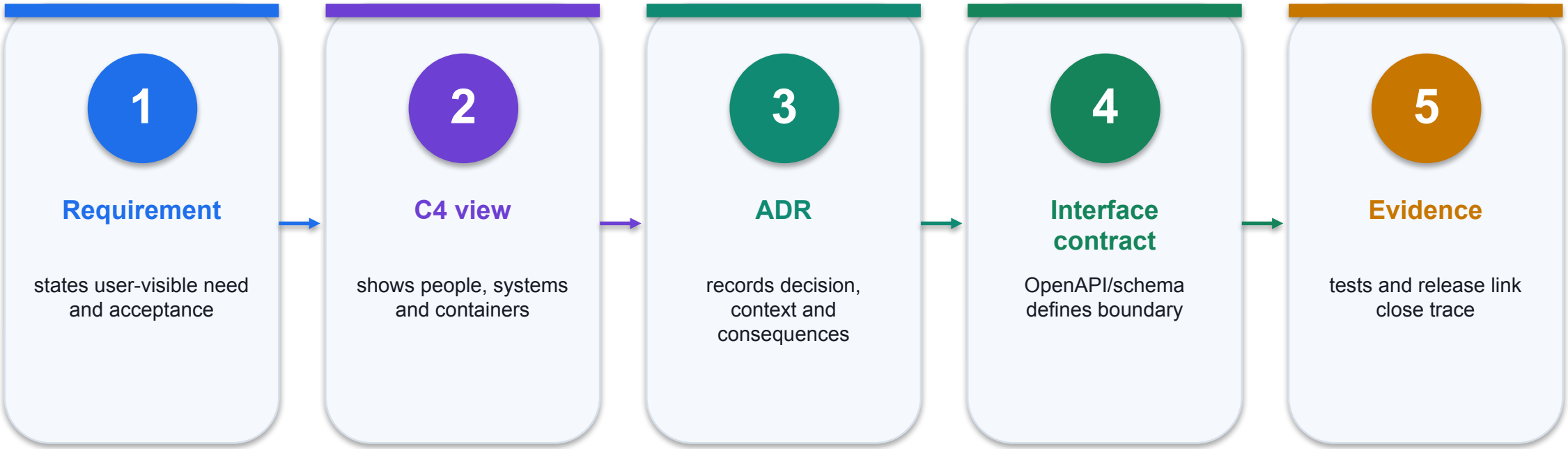
## ACCEPTANCE EVIDENCE

- ✓ Pipeline recreates build from lock file
- ✓ Artifact hash is recorded
- ✓ Smoke test proves health and create/read path

**Record URL · command · status · build SHA**


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# ADR, C4 and requirements traceability




**DESIGN RULE** Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.


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# ADR, C4 and requirements traceability · worked artifact

REVIEWED EXCERPT · .NET 10

```
# ADR-0001: Razor Pages + controller API
Status: Accepted
Decision: page-focused UI; JSON API boundary
Consequences: shared service; two HTTP surfaces.
```

1. **Requirement ID**
   REQ-001 stable identifier
2. **Decision**
   ADR-0001 accepted/rejected options
3. **Component**
   owner and responsibility
4. **Interface**
   method/path/schema/status

K5 / A5

Full procedures: Learner Guide


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TOPIC 5 · DESIGN DETAIL

# ADR, C4 and requirements traceability · interface contract

| CONTRACT ELEMENT | OPERATIONAL MEANING | REVIEW QUESTION |
| --- | --- | --- |
| Requirement ID | REQ-001 stable identifier | Where is this enforced? |
| Decision | ADR-0001 accepted/rejected options | What is observable? |
| Component | owner and responsibility | Who owns the value? |
| Interface | method/path/schema/status | What rejects invalid input? |
| Verification | test ID, result and build SHA | What must remain stable? |

**Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.**


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TOPIC 5 · DIAGNOSE + VERIFY

# ADR, C4 and requirements traceability · failure and recovery

## FAILURE SIGNAL → LIKELY CAUSE

| | |
|---|---|
| **Diagram only** | no decision rationale or consequences |
| **Orphan test** | evidence maps to no requirement |
| **Stale contract** | OpenAPI differs from deployed endpoint |
| **Missing owner** | component boundary cannot be governed |

## ACCEPTANCE EVIDENCE

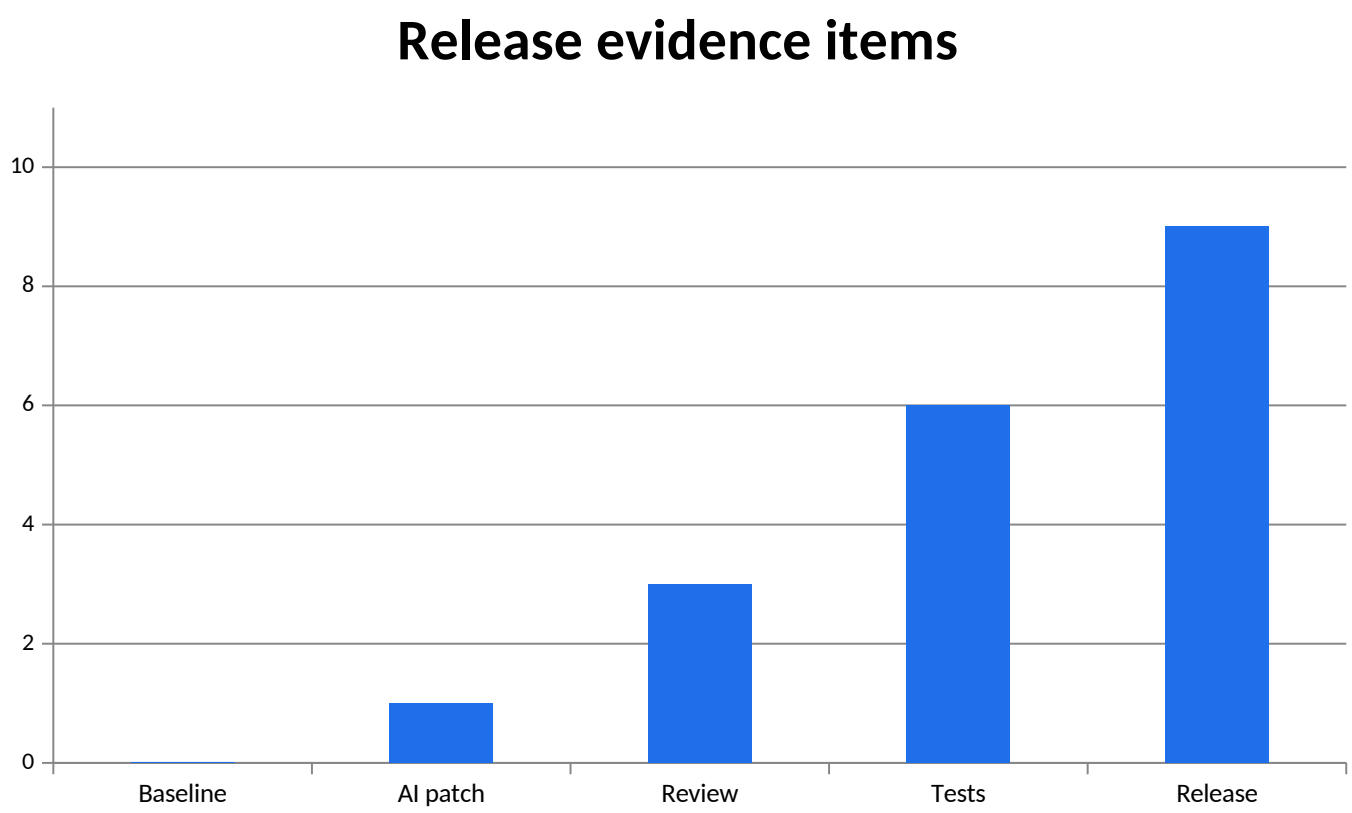
- ✓ Every LO requirement maps to component and test
- ✓ ADR names at least one rejected option
- ✓ OpenAPI hash matches released build

**Record URL · command · status · build SHA**


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TOPIC 5 · EDITABLE NATIVE CHART

# LO5 evidence profile




WHAT THE DATA SHOWS

**A deployable artifact is not a release until evidence closes the design record.**

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

EDITABLE CHART


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60 MIN

# Activity 9: Prepare an Azure-Ready Release

Create a production-neutral publish artifact with validated configuration, health endpoints and smoke evidence.




**YOU'LL PRODUCE** **Release publish directory, health checks, deployment configuration matrix and smoke-test script**

Local snapshot: activities/activity-09-production-readiness/src/LocalCRM · full commands in the Learner Guide


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# Activity 9 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Prepare LocalCRM for Azure App Service without performing a live deployment. Add validated typed options, /health/live and /health/ready, structured startup logging, and a bash smoke script. Do not add credentials, an Azure resource or a portal-only step.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 9 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: Release publish directory, health checks, deployment configuration matrix and smoke-test script.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

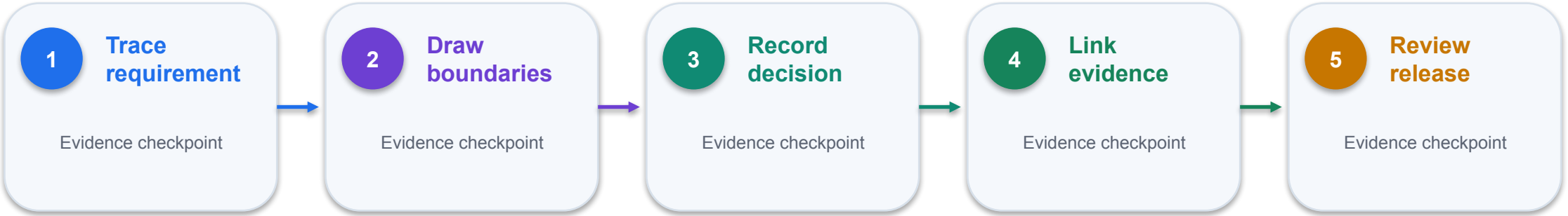
**K5/A5 EVIDENCE**


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55 MIN

# Activity 10: Produce the Traceable Design and Release Record

Map user specifications to components, interface contracts, decisions and verification evidence.




**YOU'LL PRODUCE** **C4-style diagrams, ADR, requirements traceability matrix, API contract hash and release checklist**

Local snapshot: activities/activity-10-design-documentation/src/LocalCRM · full commands in the Learner Guide


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# Activity 10 · reviewed AI contract

**VIBE SCRIPT · COPY EXACTLY**

Using requirements.md, component-map.md, openapi-v1.json and the Activity evidence, draft ADR-0001 and a traceability matrix. Every requirement must map to a component, interface or page, control, test/evidence and status. Mark gaps; do not invent evidence.

| | | |
|---|---|---|
| 1 | **Scope** | Only named files and component boundary |
| 2 | **Version** | net10.0 APIs and pinned packages |
| 3 | **Safety** | No credentials or hidden destructive step |
| 4 | **Evidence** | Build, test and HTTP acceptance named |

**The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.**


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# Activity 10 · acceptance and diagnosis

## ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: C4-style diagrams, ADR, requirements traceability matrix, API contract hash and release checklist.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

## IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

**K5/A5 EVIDENCE**


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LO5 · K5 / A5

# Topic 5 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.


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WHAT YOU CAN NOW PROVE

# Course Synthesis

1

## LO1

Determine basic software components using ASP.NET framework to meet functional specifications.

2

## LO2

Utilize ASP.NET framework to develop web apps in adherence to organizational practices.

3

## LO3

Select essential ASP.NET controls and features to meet software design requirements.

4

## LO4

Examine the interoperability and functionality of ASP.NET software components.

5

## LO5

Generate ASP.NET design documentation aligned with user specifications.


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DOWNLOAD · COMPLETE · SUBMIT

# Courseware and Assessment on the LMS

1 **Courseware**

Download the current slides and Learner Guide.

2 **Activities**

Use the repository snapshots and mock CRM data.

3 **Assessment**

Complete WA then PP in the assigned window.

4 **Submission**

Upload answers and required practical evidence.

https://lms-tms.tertiaryinfotech.com/


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FINAL REMINDER · OPEN BOOK

# Assessment

1

### WA · 16:00–17:00

Five SAQ items assess K1–K5.

2

### PP · 17:00–18:00

Five practical tasks assess A1–A5.

3

### Attendance

Complete assessment digital attendance.

4

### Submission

Upload evidence on the LMS and sign the record.


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FINAL MANDATORY SEQUENCE

# Assessment Flow




**Courseware and assessment: https://lms-tms.tertiaryinfotech.com/**


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# Digital Attendance (Mandatory)

1 **Scan**

Use the QR code shown by trainer/administrator.

2 **Submit**

Confirm the assessment attendance event.

3 **Check**

Resolve any missing record before leaving.

4 **Protect**

Do not photograph the assessment script.


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THANK YOU

# Build it. Inspect it. Prove it.

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